Organic Chemistry

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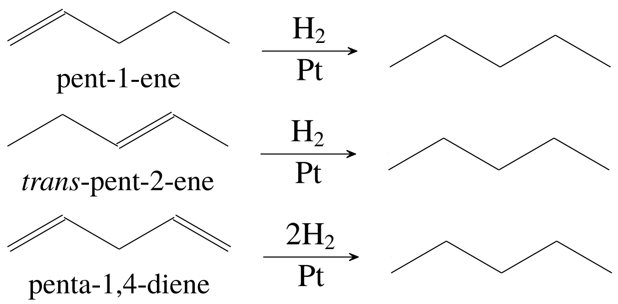
15 Conjugated Systems, Orbital Symmetry, and Ultraviolet Spectroscopy

15.1 Introduction 20260303

1. **Conjugated** double bonds: separated by one single bond.
2. **Isolated** double bonds: separated by two or more single bonds.
3. **Cummulated** double bonds: allenes, adjacent to each other.
4. Conjugated double bonds are more stable than isolated double bonds.

15.2 Stabilities of Dienes 20260303

1. Using **heats of hydrogenation** to compare the relative stabilities of alkenes. The more stable the compound, the less exothermic during hydrogenation.

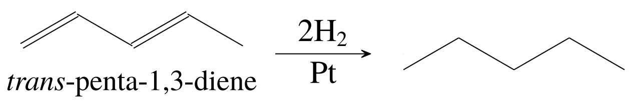


$\Delta H^\circ = -126\,\text{kJ}$

$\Delta H^\circ = -116\,\text{kJ}$

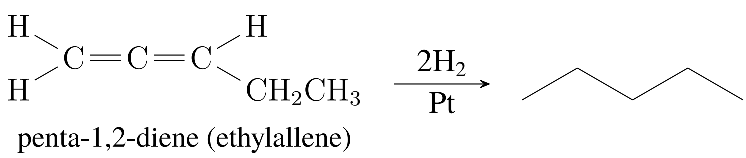
$\Delta H^\circ = -252\,\text{kJ}$

2. For conjugated dienes, the heat of hydrogenation is less than the sum for the individual double bonds, showing that the conjugated diene has extra stability.



$\Delta H^\circ = -225\,\text{kJ}$ predicted $-242\,\text{kJ}$

3. The heat of hydrogenation of penta-1,2-diene is larger than any of the other pentadienes.



$\Delta H^\circ = -292\,\text{kJ}$

4. (1) The cumulated double bonds are less stable than isolated double bonds and much less stable than conjugated double bonds.
   (2) Corresponding aklynes are less stable than alkenes.
   (3) More substitution on double bonds or triple bonds increases the stability.

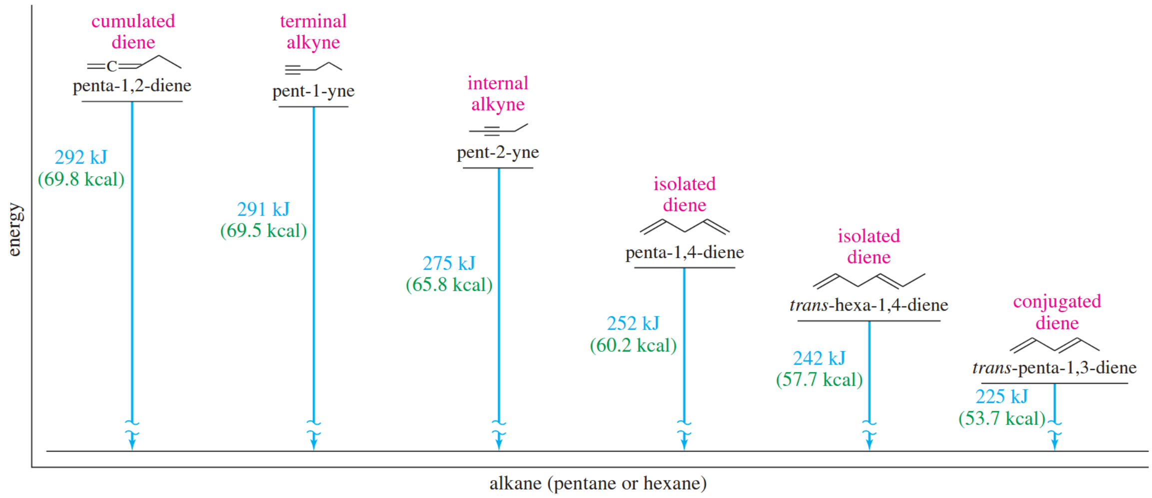


Figure 15.1 Relative Stabilities of Dienes and Alkynes

Example 15.1 Rank each group of compounds in order of increasing heat of hydrogenation.

(a) hexa-1,2-diene; hexa-1,3,5-triene; hexa-1,3-diene; hexa-1,4-diene; hexa-1,5-diene; hexa-2,4-diene

(b)

Solution to be filled with chemfig

Example 15.2 In a strongly acidic solution, cyclohexa-1,4-diene tautomerizes to cyclohexa-1,3-diene. Propose a mechanism for this rearrangement, and explain why it is energetically favorable.

Solution to be filled with chemfig

15.3 Molecular Orbital Picture of a Conjugated System 20260303 20260305

1. The extra stability in the conjugated molecule is called the **resonance energy** of the system, and can be explained by examining their **molecular orbitals**.

15.3.1 Structure and Bonding of Buta-1,3-diene

1. Figure 15.2 shows the most stable conformation of buta-1,3-diene. This conformation is planar, with the *p* orbitals on the two pi bonds aligned.
2. The $1.48\,\text{Å}$ central $\mathrm{C2-C3}$ single bond is shorter than the $1.54\,\text{Å}$ carbon-carbon bonds typical of alkanes.
3. Electrons are **delocalized** over molecules, creating some pi overlap and pi bonding in the bond.

4. To represent the bonding in conjugated systems accurately, we must consider molecular orbitals that represent the entire conjugated pi system.

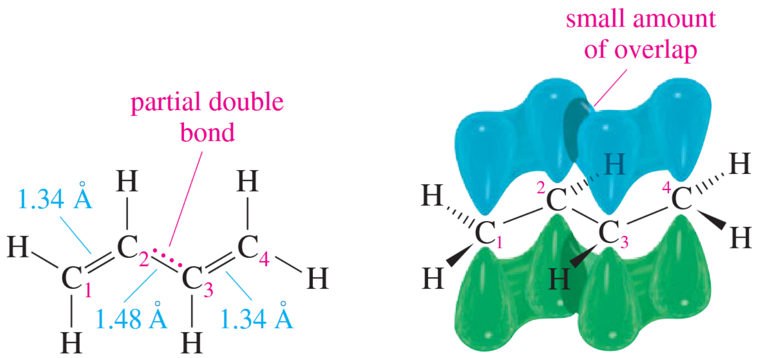


Figure 15.2 Structure of Buta-1,3-diene

15.3.2 Constructing the Molecular Orbitals of Buta-1,3-diene

1. Molecular orbitals (MOs)
   (1) Bonding, nonbonding and antibonding orbitals
   (2) The plus and minus signs indicate the *phase of the wave function*, **not electrical charges**.
   (3) **Constructive overlap**: lobes of same sign overlap, an important feature of all bonding molecular orbitals.
   (4) **Destructive overlap**: lobes of opposite sign overlap, cancelling the wave function and form a **node**.
   (5) Electrons have lower energy in the **bonding MO** than in the original p orbitals, and higher energy in the **antibonding MO**.
   (6) Stable molecules tend to have filled bonding MOs and empty antibonding MOs.

   Note Several important principles
   (1) Constructive overlap results in a bonding interaction; destructive overlap results in an antibonding interaction.
   (2) The number of MOs is always the same as the number of atomic orbitals used to form the MOs.
   (3) The MOs have energies that are symmetrically distributed above and below the energy of the starting p orbitals. Half are bonding MOs, and half are antibonding MOs.

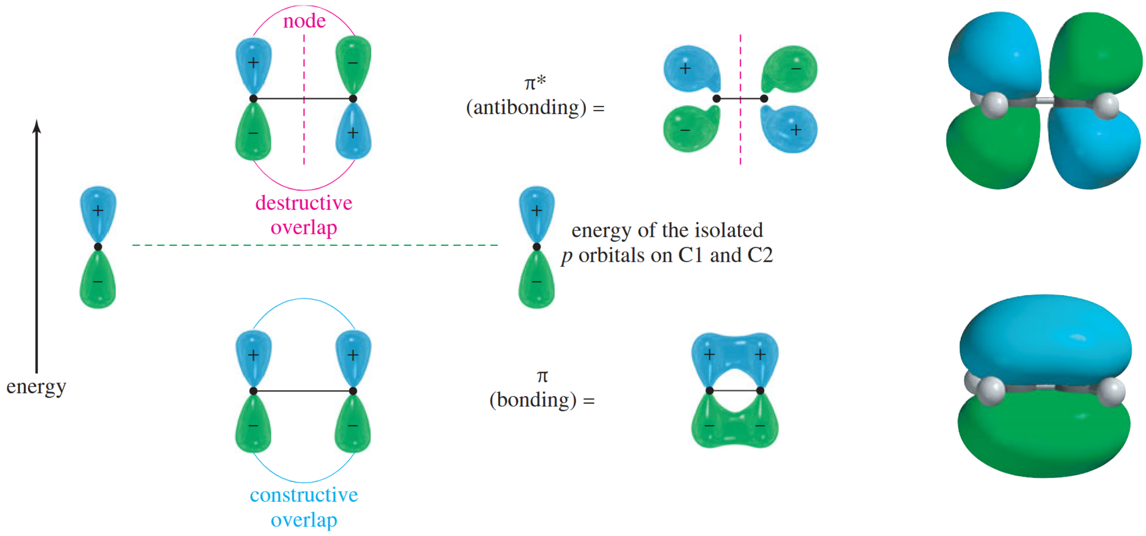


Figure 15.3 Pi Molecular Orbitals of Ethylene

2. Constructing the Molecular Orbitals of Buta-1,3-diene
   (1) Four *p* orbitals form four pi molecular orbitals, two bonding and two antibonding.
   (2) Simple straight-line representation of four orbitals.
3. π_1 MO
   (1) Lowest energy.
   (2) All bonding interactions.
   (3) Exceptionally stable: three bonding interactions, and electrons are delocalized over four nuclei.

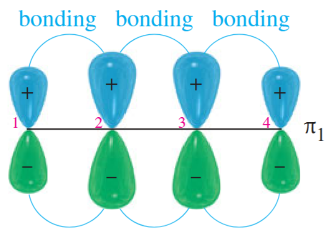


Figure 15.4 π_1 bonding MO of buta-1,3-diene

4. π_2 MO
   (1) One vertical node in the center.
   (2) Two bonding interactions and one antibonding interaction.
   (3) A bonding orbital (2 bonding $-$ 1 antibonding $=$ 1 bonding).
   (4) Higher in energy and weaker than π_1 MO.

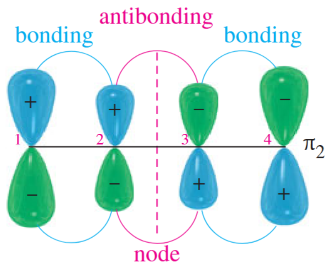


Figure 15.5 π_2 bonding MO of buta-1,3-diene

5. π_3^* MO
   (1) Two nodes.
   (2) One bonding interactions and two antibonding interaction.
6. The bonding MOs are filled and the antibonding MOs are vacant.
7. The average energy of the electrons is slightly lower in butadiene. This lower energy is the resonance stabilization of the conjugated diene.
8. Conformations of 1,3-Butadiene: *s*-trans and *s*-cis, the former, less crowded, is more stable than the latter.
9. The *s*-cis and *s*-trans conformers of butadiene (and all the skew conformations in between) easily interconvert at room temperature (The basis of DA reaction).

15.4 Allylic Cations 20260305

1. Conjugated compounds undergo a variety of reactions, many of which involve intermediates that retain some of the resonance stabilization of the conjugated system.
2. allyl group: $CH_2{=}CH{-}CH_2{-}$
3. Allylic cations and radicals are stabilized by delocalization.
4. Double bond interact with cations(exchange).
5. Stability of carbocations: $H_3C^+ < 1° < 2°$, allyl $< 3°$, substituted allylic

15.5 1,2- and 1,4-Addition to Conjugated Dienes 20260305

1. Electrophilic addition to the double bond produces the most stable intermediate.
2. For conjugated dienes, the intermediate carbocation is a resonance stabilized allylic cation.
3. MECHANISM 15-1 1,2- and 1,4-Addition to a Conjugated Diene
   (1) Protonation of one of the double bonds forms a resonance-stabilized allylic cation.
   (2) A nucleophile attacks at either electrophilic carbon atom.

15.6 Kinetic versus Thermodynamic Control in the Addition of HBr to Buta-1,3-diene 20260305

1. $-80\,^{\circ}\mathrm{C}$ 1,2-addition (kinetic control), $40\,^{\circ}\mathrm{C}$ 1,4-addition (thermodynamic control)
2. The 1,2-addition will be the faster addition at any temperature (lower E_a, secondary carbocation intermediate), but the 1,4-addition will be more stable (higher E_a, but more stable, more substituted alkene product).
3. The first step of the reaction is kenetically controlled, the second step is thermodynamically controlled.
4. Thermodynamic control at $40\,^{\circ}\mathrm{C}$
5. Kinetic control at $-80\,^{\circ}\mathrm{C}$

Example 15.3 hint: In many cases, an allylic carbocation is more stable than a bromonium ion.

15.7 Allylic Radicals 20260305

1. Allylic radicals are stabilized by resonance delocalization.
2. In a free-radical bromination of cyclohexene, substitution occurs entirely at the allylic position.
3. MECHANISM 15-2 Free-Radical Allylic Bromination
   (1) Initiation: Formation of radicals.

   Note Cleavage should happen in nonpolar solvents. e.g. CCl_4
4. Stability of free radicals: CH_3 < 1° < 2° < 3° < allylic < benzylic
5. Bromination Using N-Bromosuccinimide (NBS)

Example 15.4 prob 15-11

15.8 Molecular Orbitals of the Allylic System 20260305

15.9 Electronic Configurations of the Allyl Radical, Cation, and Anion 20260305

15.10 SN2 Displacement Reactions of Allylic Halides and Tosylates 20260305

1. Allyl bromide reacts with nucleophiles by the mechanism about 40 times faster than n-propyl bromide.

2. Allylic halides are so reactive that they couple with Grignard and organolithium reagents

15.11 The Diels–Alder Reaction 20260310

1. Otto Diels and Kurt Alder, alkenes and alkynes with electron-withdrawing groups (electron-deficient alkene, dienophile) add to conjugated dienes to form sixmembered rings (cyclohexene)

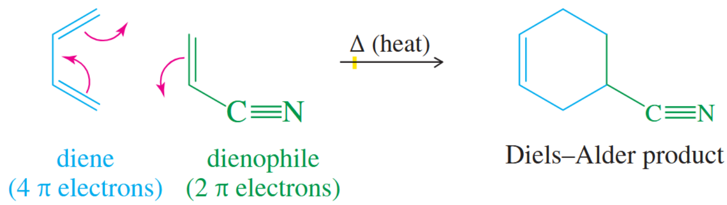


Figure 15.6 The Diels–Alder Reaction

2. a $[4+2]$ cycloaddition

Mechanism 15.1

15.11.1 Stereochemical Requirements of the Diels–AlderTransition State

1. *s*-cis Conformation of the Diene
   (1) The diene must be in the *s*-cis conformation to react.
   (2) Dienes with functional groups that hinder the *s*-cis conformation react more slowly than butadiene. Dienes with functional groups that hinder the *s*-trans conformation react faster than butadiene.
2. syn Stereochemistry
   (1) There is no opportunity for any of the substituents to change their stereochemical positions during the course of the reaction.
3. The Endo Rule
   (1) Dienophile has a pi bond in its electron-withdrawing group (as in a carbonyl group or a cyano group).
   (2) Secondary overlap: an overlap of the *p* orbitals of the electron-withdrawing group with the *p* orbitals of C2 and C3 of the diene.
   (3) Secondary overlap helps to stabilize the transition state.
   (4) This position is called the endo position, and This stereochemical preference for the electron-withdrawing substituent to appear in the endo position is called the endo rule.

15.11.2 Diels–Alder Reactions Using Unsymmetrical Reagents

1. When the diene and dienophile are both unsymmetrically substituted, the DA reaction usually gives a single product (or a major product).
2. Predict the major product by considering how the substituents polarize the diene and the dienophile in their charge-separated resonance forms.
3. An electrondonating substituent (D) on the diene and an electron-withdrawing substituent (W) on the dienophile usually show either a 1,2- or 1,4-relationship in the product.

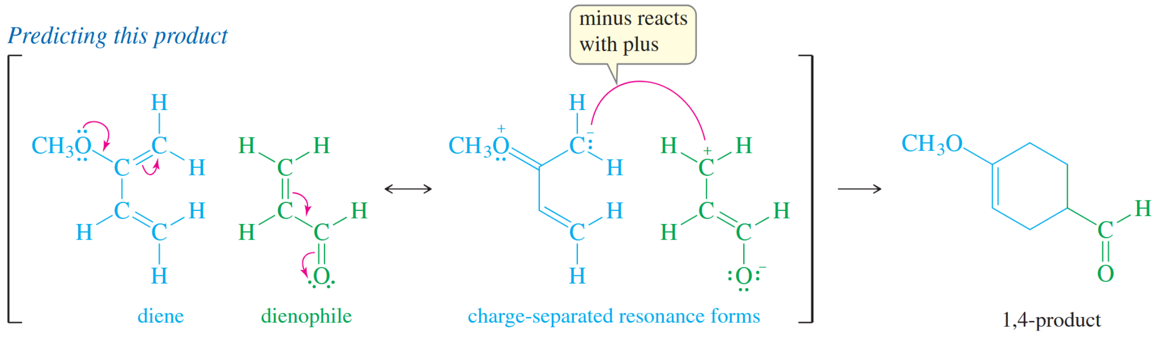


Figure 15.7 Formation of 1,4-product

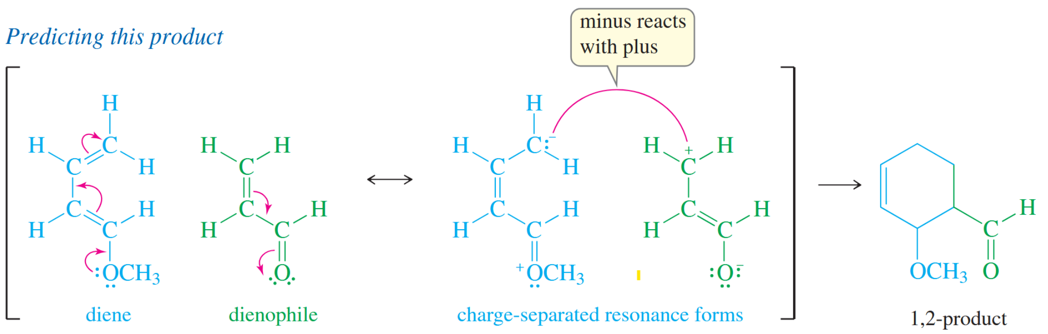


Figure 15.8 Formation of 1,2-product

4. Remember the major product will not be a 1,3-product.

15.12 The Diels–Alder as an Example of a Pericyclic Reaction 20260310 20260312

1. Pericyclic reactions involve the concerted forming and breaking of bonds within a closed ring of interacting orbitals.
2. Conservation of orbital symmetry: the molecular orbitals of the reactants must flow smoothly into the MOs of the products without any drastic changes in symmetry.

15.12.1 Conservation of Orbital Symmetry in the Diels–Alder Reaction

1. Highest Occupied Molecular Orbital (HOMO), Lowest Unoccupied Molecular Orbital (LUMO)
2. The HOMO of butadiene has the correct symmetry (plus with plus and minus with minus) to overlap in phase with the LUMO of ethylene.
3. The bonding interactions stabilize the transition state and promote the concerted reaction. This favorable result predicts that the reaction is symmetry-allowed.

15.12.2 The "Forbidden" [2+2] Cycloaddition

1.

15.12.3 Photochemical Induction of Cycloadditions

1. This higher orbital, formerly the LUMO, is now occupied: It is the new the HOMO*, the HOMO of the excited molecule.

Example 15.5 Show that the [4 + 2] Diels–Alder reaction is photochemically forbidden. (In basic chemistry not react, but may react in different symmetry gouuxiang)

15.13 Ultraviolet Absorption Spectroscopy 20260312

1. Ultraviolet (UV) spectroscopy detects the electronic transitions of **conjugated systems** and provides information about the length and structure of the conjugated part of a molecule
2. UV spectroscopygiving more specialized information than does IR or NMR.

15.13.1 Spectral Region

1. jadsfkjas

15.13.2 Ultraviolet Light and Electronic Transitions

1. A photon with the right amount of energy can excite an electron from the bonding orbital (π) to the antibonding orbital ($\pi*$). This transition from a bonding orbital to a antibonding orbital is called a $\pi \to \pi*$ transition.
2. In conjugated systems, there are electronic transitions with lower energies that correspond to wavelengths longer than $200\,\mathrm{nm}$.

3. In general, the energy difference between HOMO and LUMO decreases as the length of conjugation increases.

15.13.3 Obtaining an Ultraviolet Spectrum

1. The sample solution is placed in a quartz cell, and some of the solvent is placed in a reference cell.
2. Compare the amount of light transmitted through the sample (the sample beam) with the amount of light in the reference beam. Measure the intensity ratio of the reference (I_r) beam compared with the sample beam (I_s).
3. The absorbance, A, of the sample at a particular wavelength is governed by *Beer's law*.

$$A = \log \frac{I_\mathrm{r}}{I_\mathrm{s}} = \epsilon c l$$

4. Molar absorptivity (ϵ) is a measure of how strongly the sample absorbs light at that wavelength.
5. The molar absorptivity of onjugated system is about 5000 to 50000.

15.13.4 Interpreting UV–Visible Spectra

1. buta-1,3-diene $217\,\mathrm{nm}$, cyclohexa-1,3-diene $256\,\mathrm{nm}$.
2. The Woodward-Fieser Rules
3. the values of increase by about $30\,\mathrm{nm}$ to $40\,\mathrm{nm}$ for each double bond extending the conjugated system. Alkyl groups also increase the value of by about $5\,\mathrm{nm}$ per alkyl group.

Example 15.6

end of Ch 15

16 Aromatic Compounds

16.1 Introduction: The Discovery of Benzene 20260312 20260317

1.

16.2 The Structure and Properties of Benzene 20260312 20260317

1. Kekulé structure, failed to explain existence of only one isomer of 1,2-dichlorobenzene.
2. The Resonance Representation: a resonance hybrid between the two Kekulé structures, represented by drawing a circle inside the six-membered ring as a combined representation.
3. Structure of Benzene: Each sp^2 hybridized C in the ring has an unhybridized p orbital perpendicular to the ring which overlaps around the ring.
4. An aromatic compound is a cyclic compound containing some number of conjugated double bonds and having an unusually large resonance energy.
5. The Unusual Reactions of Benzene: will not undergo the typical reactions of polyenes.
6. **Unusual Addition of Bromine to Benzene**
   (1) When bromine adds to benzene, a catalyst such as $\mathbf{FeBr_3}$ is needed.
   (2) Substitution of a hydrogen by bromine.
   (3) No addition of Br2 to the double bond.
7. The difference between the predicted and the observed value of hydrogenation is called the resonance energy.
8. Annulenes
   (1) Annulenes are hydrocarbons with alternating single and double bonds.
   (2) Cyclobutadiene is [4]annulene, Benzene is [6]annulene Ccyclooctatetraene is [8]annulene.
   (3) Reactivity: Cyclobutadiene is so reactive that it dimerizes before it can be isolated. Cyclooctatetraene adds Br_2 readily to the double bonds.
   (4) tub conformation of cyclooctatetraene

16.3 The Molecular Orbitals of Benzene 20260317

1. Six overlapping p orbitals must form six molecular orbitals.
   Three will be bonding, three antibonding.
   Lowest energy MO will have all bonding interactions, no nodes.
   As energy of MO increases, the number of nodes increases.

2. First MO of Benzene
   Intermediate MO of Benzene
3. **Electronic Energy Diagram** for Benzene: Six electrons fill three bonding pi orbitals. All bonding orbitals are filled ("closed shell"), an extremely stable arrangement.

16.4 The Molecular Orbital Picture of Cyclobutadiene 20260317

1. Electronic Energy Diagram for Cyclobutadiene: Following Hund's rule, two electrons are in separate nonbonding molecular orbitals. This diradical would be very reactive.
2. The Polygon Rule: The molecular orbital energy diagram of a regular, completely conjugated cyclic system has the same polygonal shape as the compound, with one vertex (the all-bonding MO) at the bottom.
3. However, cyclooctatetraene will not follow the Polygon Rule (planar) but will form a tub conformation to be more stable.

16.5 Aromatic, Antiaromatic, and Nonaromatic Compounds 20260317

1. Aromatic compounds are those that meet the following criteria:
   (1) Structure must be cyclic with conjugated pi bonds.
   (2) Each atom in the ring must have an unhybridized *p* orbital (sp^2 or *sp*).
   (3) The *p* orbitals must overlap continuously **around the ring**. Structure must be planar (or close to planar for effective overlap to occur).
   (4) Delocalization of the pi electrons over the ring must lower the electronic energy.
2. An antiaromatic compound is one that meets the first three criteria, but delocalization of the pi electrons over the ring increases the electronic energy. Compounds or ions.
3. Nonaromatic compounds do not have a continuous ring of overlapping *p* orbitals and may be nonplanar.

16.6 Hückel's Rule 20260317

1. To qualify as aromatic or antiaromatic, a cyclic compound must have a continuous ring of overlapping p orbitals, usually in a planar conformation.
2. Once the criteria are met, Hückel's rule applies:
   If the number of pi electrons in the cyclic system is: the system is aromatic. (4N) the system is antiaromatic. N is an integer, commonly 0, 1, 2, or 3.
3. 4N Annulenes

   (1) [4]annulene is antiaromatic.
   (2) [8]annulene would be antiaromatic, but it's not planar, so it's nonaromatic.
   (3) Larger 4N annulenes are not antiaromatic because they are flexible enough to become nonplanar.
4. 4N+2 Annulenes
   (1) [6]annulene (benzene) is aromatic.
   (2) [10]annulene is aromatic except for the isomers that are nonplanar. (two hydrogen atoms interfere with each other)
   (3) Some larger 4N+2 annulenes can achieve planar conformation, they are aromatic.

16.7 MO Derivation of Hückel's Rule 20260317

1. Aromatic compounds have (4N+2) electrons and the orbitals are filled.
2. Antiaromatic compounds have only 4N electrons and has unpaired electrons in two degenerate orbitals.

16.8 Aromatic ions 20260317 20260319

16.8.1 Cyclopentadienyl Ions

1. The cation has an empty p orbital, 4 electrons, so it is antiaromatic.
2. The anion has a nonbonding pair of electrons in a *p* orbital, 6 electrons, it is aromatic.
3. Cyclopentadiene is acidic because deprotonation will convert it to an aromatic ion.
4. Deprotonation of Cyclopentadiene:
   By deprotonating the sp^3 carbon of cyclopentadiene, the electrons in the p orbitals can be delocalized over all five carbon atoms and the compound would be aromatic.
   Cyclopentadiene is acidic because deprotonation will convert it to an aromatic ion.
5. Resonance Forms of Cyclopentadienyl Ions
   The resonance picture gives **a misleading suggestion** of stability.

16.8.2 The Cycloheptatrienyl Ions

1. The cycloheptatrienyl cation has 6 pi electrons and an empty *p* orbital.
2. The cycloheptatrienyl cation is easily formed by treating the corresponding alcohol with dilute (0.01 molar) aqueous sulfuric acid.
3. The cycloheptatrienyl cation is called the tropylium ion. This aromatic ion is much less reactive than most carbocations.

16.8.3 The Cyclooctatetraene Dianion

1. Cyclooctatetraene reacts with potassium metal to form an aromatic dianion.
2. The dianion has 10 pi electrons and is aromatic.
3. Biphenyl has the following structure. p752
4. Hexahelicene p754

Example 16.1 Explain why each compound or ion should be aromatic, antiaromatic, or nonaromatic.

1. the cyclononatetraene cation, anti
2. the cyclononatetraene anion, aro
3. the [16]annulene dianion, aro if planar
4. the [18]annulene dianion, anti
5. non
6. the [20]annulene dication, aro if planar

Example 16.2 The following hydrocarbon has an unusually large dipole moment. Explain how a large dipole moment might arise.

SOLUTION The reason for the dipole can be seen in a resonance form distributing the electrons to give each ring 6 pi electrons. This resonance picture gives one ring a negative charge and the other ring a positive charge.

Example 16.3 When 3-chlorocyclopropene is treated with $AgBF_4$, AgCl precipitates. The organic product can be obtained as a crystalline material, soluble in polar solvents such as nitromethane but insoluble in hexane. When the crystalline material is dissolved in nitromethane containing KCl, the original 3-chlorocyclopropene is regenerated. Determine the structure of the crystalline material, and write equations for its formation and its reaction with chloride ion.

Example 16.4 The polarization of a carbonyl group can be represented by a pair of resonance structures:

16.8.4 Summary of Annulenes and Their Ions

1. The 2, 6, and 10 electron systems are aromatic, while the 4 and 8 electron systems are antiaromatic if they are planar.

16.9 Heterocyclic Aromatic Compounds 20260319

16.9.1 Pyridine

1. Pyridine has a nitrogen atom in place of one of the six $\mathrm{C{-}H}$ units of benzene.
2. Pyridine has a six-membered heterocyclic ring with six pi electrons.
3. The nonbonding electrons are in an sp^2 hybrid orbital in the plane of the ring, perpendicular to the pi system and do not overlap with it.
4. Pyridine is basic because of its available pair of nonbonding electrons.
5. The pyridinium ion is still aromatic because the additional proton has no effect on the electrons of the aromatic sextet.

16.9.2 Pyrrole

1. Pyrrole is an aromatic five-membered heterocycle, with one nitrogen atom and two double bonds.
2. The pyrrole nitrogen atom is sp^2 hybridized, and has a lone pair of electrons in the unhybridized p orbital.
3. The unhybridized orbital overlaps with the p orbitals of the carbon atoms to form a continuous ring.
4. Pyrrole is a much weaker base than pyridine. To form a bond to a proton requires the use of one of the electron pairs in the aromatic sextet.
5. Basicity: Delocalized lone pair of electrons is not basic, usually with a hydrogen atom on the nitrogen atom.

16.9.3 Pyrimidine and Imidazole

1. Pyrimidine is a six-membered heterocycle with two nitrogen atoms situated in a 1,3- arrangement.
2. Imidazole is an aromatic five-membered heterocycle with two nitrogen atoms.
3. Purine has an imidazole ring fused to a pyrimidine ring. Purine has three basic nitrogen atoms and one pyrrole-like nitrogen.

Example 16.5 The proton NMR spectrum of 2-pyridone gives the chemical shifts shown.

16.9.4 Furan and Thiophene 20260319

1. Furan is an aromatic five-membered heterocycle, and the heteroatom is oxygen.
2. Thiophene is similar to furan, with a sulfur atom in place of the furan oxygen. The sulfur atom uses an unhybridized $3p$ orbital to overlap with the $2p$ orbitals on the carbon atoms.

Example 16.6 Explain why each compound is aromatic, antiaromatic, or nonaromatic.

Example 16.7 Borazole, $B_3N_3H_6$, is an unusually stable cyclic compound. Propose a structure for borazole, and explain why it is aromatic.

16.10 Polynuclear Aromatic Hydrocarbons 20260324

1. Composed of two or more fused benzene rings, sharing two carbon atoms and the bond between them.
2. Naphthalene: the simplest fused aromatic compound, consisting of two fused benzene rings.
3. Anthracene and Phenanthrene: As the number of fused aromatic rings increases, the resonance energy per ring continues to decrease and the compounds become more reactive.
4. Larger Polynuclear Aromatic Hydrocarbons: have more fused rings than anthracene and phenanthrene, and they have less resonance energy per ring and are more reactive.

16.11 Aromatic Allotropes of Carbon 20260324

16.11.1 Allotropes of Carbon: Diamond

1. Amorphous carbon refers to charcoal, soot, coal, and carbon black.
2. Diamond has a crystalline structure containing tetrahedral carbon atoms linked together in a three-dimensional lattice.

16.11.2 Graphite

1. Graphite has the layered planar structure.
2. Each layer of graphite is a nearly infinite lattice of fused aromatic rings. No bonds but van der Waals forces hold the layers together.

16.11.3 Fullerenes

16.12 Fused Heterocyclic Compounds 20260324

Electrostatic potential map shows electron-rich (red) and electron-poor (blue).

16.13 Nomenclature of Benzene Derivatives 20260324

1. The following compounds are usually called by their historical common names, and almost never by the systematic IUPAC names:
   phenol (benzenol), toluene (methylbenzene), aniline (benzenamine), anisole (methoxybenzene), styrene (vinylbenzene), acetophenone (methyl phenyl ketone), benzaldehyde, benzoic acid
2. Many compounds are named as derivatives of benzene.
   tert-butylbenzene, nitrobenzene, ethynylbenzene (phenylacetylene), benzenesulfonic acid
3. Disubstituted benzenes are named using the prefixes **ortho-** (1,2), **meta-** (1,3), and **para-** (1,4) to specify the substitution patterns. These terms are abbreviated *o*-, *m*-, and *p*-.
4. With three or more substituents on the benzene ring, numbers are used to indicate their positions, and give the lowest possible numbers to the substituents. The carbon atom bearing the functional group that defines the base name.
5. Many disubstituted benzenes (and polysubstituted benzenes) have historical names.
   m-xylene,
6. When the benzene ring is named as a substituent on another molecule, it is called a **phenyl group**.

16.14 Physical Properties of Benzene and Its Derivatives 20260324

16.15 Spectroscopy of Aromatic Compounds 20260324 20260326

16.15.1 Infrared Spectroscopy

1. Aromatic compounds are readily identified by their infrared spectra because they show a characteristic $\mathrm{C{=}C}$ stretch around $1600\,\mathrm{cm}^{-1}$.
2. s: strong, b: broad, m: medium
3. 4000 - 2500 ($\mathrm{X{-}H}$, $\mathrm{O{-}H}$, $\mathrm{N{-}H}$, $\mathrm{C{-}H}$),
   2500 - 2000 ($\mathrm{C{\equiv}C}$, $\mathrm{C{\equiv}N}$),
   2000 - 1500 ($\mathrm{C{=}C}$, $\mathrm{C{=}O}$),
   1500 - 400 (fingerprint)

16.15.2 NMR Spectroscopy

16.15.3 Mass Spectrometry

1. The benzyl cation may rearrange to give the aromatic tropylium ion.
2. Alkylbenzenes frequently give ions corresponding to the tropylium ion at m/z 91.

16.15.4 Ultraviolet Spectroscopy

1. The most characteristic part of the spectrum is the band centered at $254\,\mathrm{nm}$, called the **benzenoid band**.
2. About three to six small, sharp peaks (called fine structure) usually appear in this band.

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17 Reactions of Aromatic Compounds

17.1 Electrophilic Aromatic Substitution 20260326

1. Although benzene's pi electrons are in a stable aromatic system, they are available to attack a strong electrophile to give a carbocation.
2. This resonance-stabilized carbocation is called a **sigma complex** because the electrophile is joined to the benzene ring by a new sigma bond.
3. The sigma complex (also called an arenium ion) is not aromatic because the sp^3 hybrid carbon atom interrupts the ring of p orbitals.

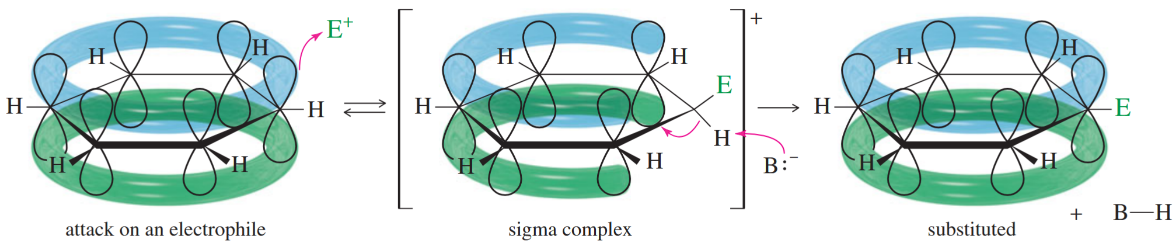


Figure 17.1 Sigma Complex

Mechanism 17.1 Electrophilic Aromatic Substitution

- Step 1: Attack on the electrophile forms the sigma complex.
- Step 2: Loss of a proton regains aromaticity and gives the substitution product.

Example 17.1 Iodination of toluene

17.2 Halogenation of Benzene 20260326

17.2.1 Bromination of Benzene

1. Bromination follows the general mechanism for electrophilic aromatic substitution.
2. Bromine itself is not sufficiently electrophilic, and the formation of Br^+ is difficult.
3. A strong Lewis acid such as catalyzes the reaction by forming a complex with Br_2 that reacts like Br^+.

Mechanism 17.2 Bromination of Benzene

17.2.2 Chlorination of Benzene

1. Chlorination of benzene works much like bromination, except that aluminum chloride ($AlCl_3$) is most often used as the Lewis acid catalyst.

17.2.3 Iodination of Benzene

1. Iodination of benzene requires an acidic oxidizing agent, such as nitric acid, to produce iodide cation.

Example 17.2 flourination

17.3 Nitration of Benzene 20260326

1. Benzene reacts with hot, concentrated nitric acid to give nitrobenzene. A safer and more convenient procedure uses a mixture of nitric acid and sulfuric acid.
2. Sulfuric acid is a catalyst, allowing nitration to take place more rapidly and at lower temperatures.

Mechanism 17.3 Nitration of Benzene

- Preliminary steps: Formation of the nitronium ion, NO_2^+.
- Step 1: Attack on the electrophile forms the sigma complex.
- Step 2: Loss of a proton gives nitrobenzene.

3. Aromatic nitro groups are easily reduced to amino ($—NH_2$) groups by treatment with an active metal such as tin, zinc, or iron in dilute acid.
4. Nitration followed by reduction is often the best method for adding an amino group to an aromatic ring.

17.4 Sulfonation of Benzene 20260326

17.4.1 Sulfonation

1. Sulfur trioxide (SO_3) is the electrophile.
2. “Fuming sulfuric acid” is the common name for a solution of 7% SO_3 in H_2SO_4.

Mechanism 17.4 Sulfonation of Benzene

17.4.2 Desulfonation

1. A sulfonic acid group may be removed from an aromatic ring by heating in dilute sulfuric acid.
2. Steam is often used as a source of both water and heat for desulfonation.

17.4.3 Protonation of the Aromatic Ring; Hydrogen–Deuterium Exchange

17.5 Nitration of Toluene: The Effect of Alkyl Substitution 20260326

1. Toluene (methylbenzene) reacts with a mixture of nitric and sulfuric acids much like benzene does, but with some interesting differences:
   (1) Toluene reacts about 25 times faster than benzene under the same conditions. Toluene is activated toward electrophilic aromatic substitution, and the methyl group is an **activating group**.
   (2) Nitration of toluene gives a mixture of products, primarily those resulting from substitution at the ortho and para positions. The methyl group of toluene is an **ortho, para-director**.
2. In ortho or para substitution of toluene, the positive charge is spread over two secondary carbons and one tertiary (3°) carbon (bearing the CH_3 group).
3. The sigma complex for meta substitution has its positive charge spread over three 2° carbons.
4. Inductive effect:
   - Electron-withdrawing: $-F$, $-Cl$, $-Br$, $-NO_2$, $-O$, $-N$
   - Electron-donating: $-CH_3$, $-R$

   Resonance effect:
   - Electron-withdrawing: $-NO_2$, $-C\equiv N$, $-C=O$
   - Electron-donating: $-OR$, $-NR_2$, $-Cl$, $-Br$

17.6 Activating, Ortho, Para-Directing Substituents 20260331

17.6.1 Alkyl Groups

1. An alkyl group is an activating substituent, and it is ortho, para-directing.
2. Substitution ortho or para to the alkyl group gives a transition state and an intermediate with the positive charge shared by the tertiary carbon atom.
3. This effect is called inductive stabilization because the alkyl group donates electron density

through the sigma bond joining it with the benzene ring.
4.

Example 17.3 Styrene (vinylbenzene) undergoes electrophilic aromatic substitution much faster than benzene, and the products are found to be primarily ortho- and para-substituted styrenes. A phenyl substituent should be ortho, para-directing like vinyl.

17.6.2 Substituents with Nonbonding Electrons

1. Alkoxyl Groups
   (1) Anisole (methoxybenzene) undergoes nitration about 10,000 times faster than benzene.
   (2) The methoxy group of anisole preferentially activates the ortho and para positions.
   (3) The nonbonding electrons of an oxygen atom adjacent to a carbocation stabilize the positive charge through resonance.
   (4) This type of stabilization is called **resonance stabilization**, and the oxygen atom is called **resonance-donating** or **pi-donating**.
   (5) The resonance effect is much stronger than the inductive effect.
   (6) A methoxy group is so strongly activating that anisole quickly brominates in water without a catalyst. In the presence of excess bromine, this reaction proceeds to the tribromide. This reaction can be controlled in low temerature and with a limited amount of bromine to give the single or double substituted product.
2. Amine Groups
   (1) A nitrogen atom with a nonbonding pair of electrons serves as a powerful ortho, para director and activating group.
   (2) Aniline undergoes a fast bromination (without a catalyst) in bromine water to give the tribromide.
   (3) Sodium bicarbonate is added to neutralize the HBr formed and to prevent protonation of the basic amino ($-NH_2$) group, which would be a electron-withdrawing, meta-directing substituent.
   (4) Nitrogen's nonbonding electrons provide resonance stabilization to the sigma complex if attack takes place ortho or para to the position of the nitrogen atom.
3. Any substituent with a lone pair of electrons on the atom bonded to the ring can provide resonance stabilization to a sigma complex.

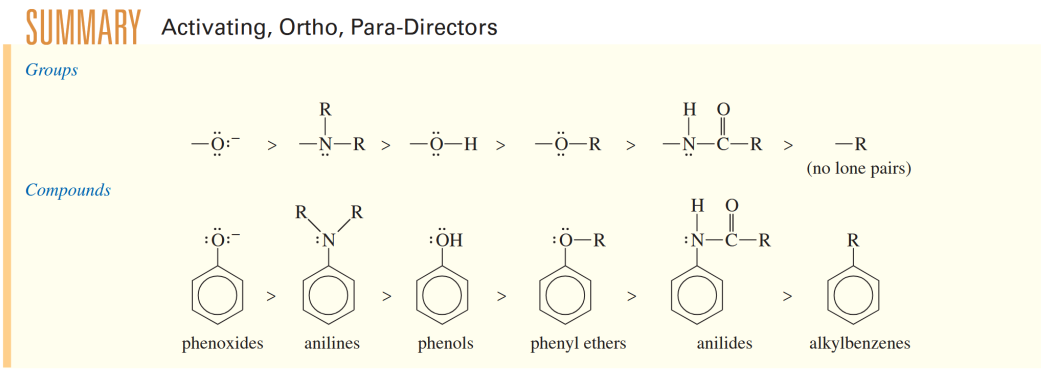


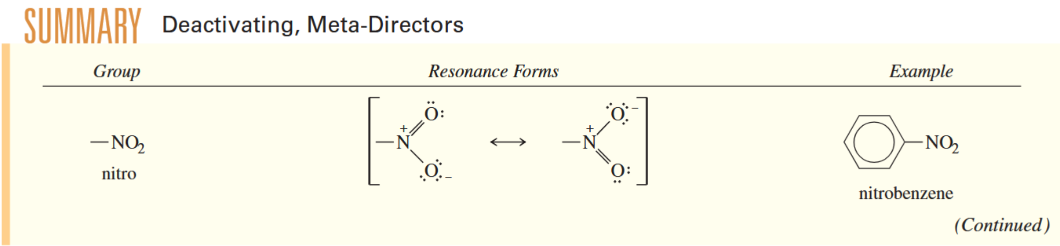
Figure 17.2 Activating, Ortho, Para-Directors

17.7 Deactivating, Meta-Directing Substituents 20260331

1. Nitrobenzene is about 100,000 times less reactive than benzene toward electrophilic aromatic substitution.
2. An electron-donating substituent activates primarily the ortho and para positions, and an electron-withdrawing substituent (such as a nitro group) deactivates primarily the ortho and para positions.
3. **Meta-directors**, often called meta-allowing substituents, deactivate the meta position less than the ortho and para positions, allowing meta substitution.
4. Nitro group is a strong deactivating group by considering its resonance forms.
5. deactivating substituents are groups with a positive charge (or a partial positive charge) on the atom bonded to the aromatic ring. As we saw with the nitro group, this positively charged atom repels any positive charge on the adjacent carbon atom of the ring. Of the possible sigma complexes, only the one corresponding to meta substitution avoids putting a positive charge on this ring carbon.

17.8 Halogen Substituents: Deactivating, but Ortho, Para-Directing 20260331

1. The halobenzenes are exceptions to the general rules. Halogens are deactivating groups, yet they are ortho, para-directors.
2. The unusual combination of properties can be explained by considering that
   (1) the halogens are strongly electronegative, withdrawing electron density from a carbon atom through the sigma bond (inductive withdrawal).
   (2) the halogens have nonbonding electrons that can donate electron density through pi bonding (resonance donation).
3.



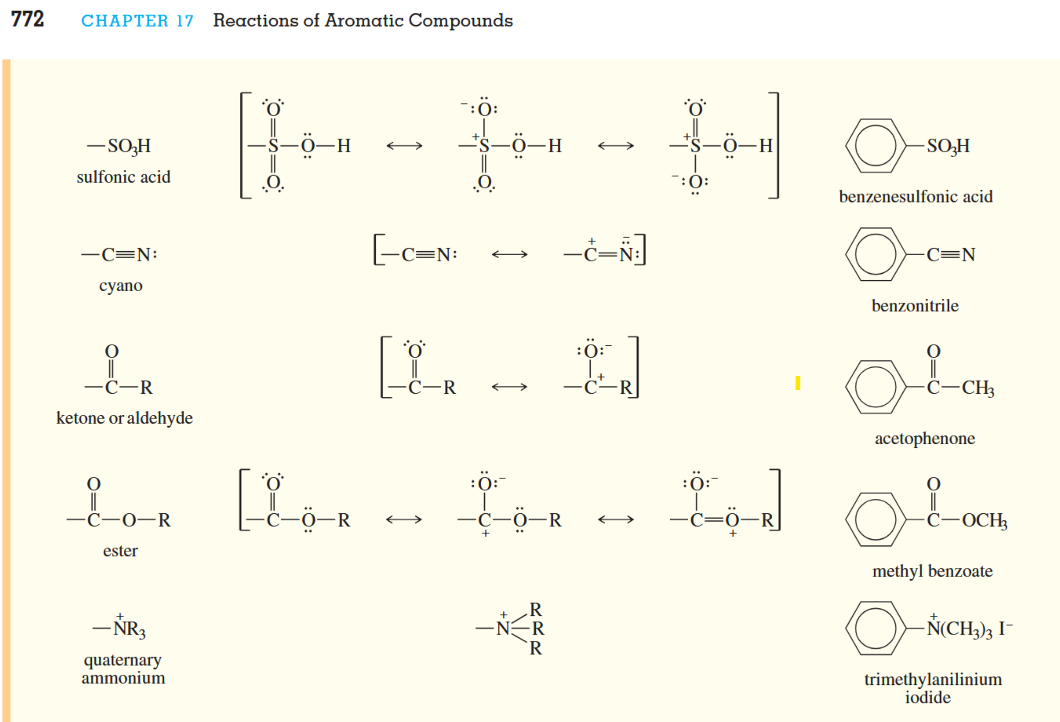


Figure 17.3 Deactivating, Meta-Directors

17.9 Effects of Multiple Substituents on Electrophilic Aromatic Substitution 20260331

1. If the groups reinforce each other, the result is easy to predict.

Example 17.4 Predict the mononitration products of the following compounds.

(a) *o*-nitrotoluene

(b) *m*-chlorotoluene

(c) *o*-bromobenzoic acid

(d) *p*-methoxybenzoic acid

(e) *m*-cresol (*m*-methylphenol)

(f) *o*-hydroxyacetophenone

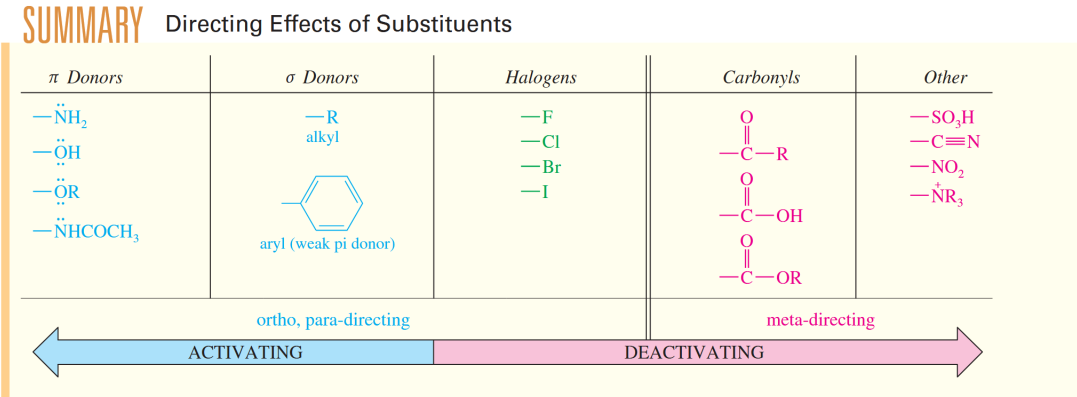


Figure 17.4 Directing Effects of Substituents

2. When the directing effects of two or more substituents conflict, it is more difficult to predict where an electrophile will react. In many cases, mixtures result.
3. When there is a conflict between an activating group and a deactivating group, the activating group usually directs the substitution. In other words, activating groups are usually stronger directors than deactivating groups.
4. Separation of substituents into three classes, from strongest to weakest.
   (1) Powerful ortho, para-directors that stabilize the sigma complexes through resonance.
   (2) Moderate ortho, para-directors
   (3) All meta-directors.

$$\text{—OH, —OR, —NR}_2 \quad > \quad \text{—R, —X} \quad > \quad \text{—}\overset{\displaystyle O}{\overset{\|}{\text{C}}}\text{—R, —SO}_3\text{H, —NO}_2$$

Figure 17.5 Substituents Strength

5. If two substituents direct an incoming electrophile toward different reaction sites, the substituent in the stronger class predominates. If both are in the same class, mixtures are likely.

Example 17.5 Predict the mononitration products of the following aromatic compounds.

(a) *p*-methylanisole
(b) *m*-nitrochlorobenzene
(c) *p*-chlorophenol
(d) *m*-nitroanisole

Example 17.6 *Biphenyl* is two benzene rings joined by a single bond. The site of substitution for a biphenyl is determined by (1) which phenyl ring is more activated (or less deactivated), and (2) which position on that ring is most reactive, using the fact that a phenyl substituent is activating and ortho, para-directing.

(a) Use resonance forms of a sigma complex to show why a phenyl substituent should be ortho, para-directing.

(b) Predict the mononitration products of the following compounds.

17.10 The Friedel–Crafts Alkylation 20260331 20260402

1. In the presence of Lewis acid catalysts such as aluminum chloride ($AlCl_3$) or ferric chloride ($FeCl_3$), alkyl halides were found to alkylate benzene to give alkylbenzenes. This useful reaction is called the **Friedel–Crafts alkylation**.

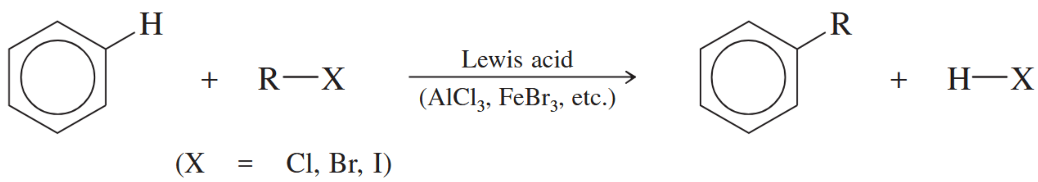


Figure 17.6 Friedel–Crafts alkylation

Mechanism 17.5 Friedel–Crafts Alkylation

Example 17.7 Propose products (if any) and mechanisms for the following $AlCl_3$-catalyzed reactions:

(a) chlorocyclohexane with benzene

(b) methyl chloride with anisole

(c) 3-chloro-2,2-dimethylbutane with isopropylbenzene

SOLUTION (c) rearrangement

2. Friedel–Crafts Alkylation Using Other Carbocation Sources
   (1) Alkenes are protonated by HF to give carbocations.
   (2) Alcohols commonly form carbocations when treated with Lewis acids such as boron trifluoride (BF_3).
3. Limitations of the Friedel–Crafts Alkylation
   (1) Friedel–Crafts reactions work only with **benzene**, **activated benzene derivatives**, and **halobenzenes**. They fail with strongly deactivated systems such as nitrobenzene, benzenesulfonic acid, and phenyl ketones.
   We can get around this limitation by adding the deactivating group or changing an activating group into a deactivating group *after* the Friedel–Crafts step.

(2) The Friedel–Crafts alkylation is susceptible to carbocation rearrangements. Only certain alkylbenzenes can be made using the Friedel–Crafts alkylation.
(3) Because alkyl groups are activating substituents, the product of the Friedel–Crafts alkylation is *more reactive* than the starting material. **Multiple alkylations** are hard to avoid.
The problem of overalkylation can be minimized by using a large excess of benzene.

Example 17.8 Devise a synthesis of *p*-nitro-*tert*-butylbenzene from benzene.

Example 17.9 Predict the products (if any) of the following reactions.

(a) (excess) bezene + isobutyl chloride + $AlCl_3$

Example 17.10 Which reactions will produce the desired product in good yield? You may assume that aluminum chloride is added as a catalyst in each case. For the reactions that will not give a good yield of the desired product, predict the major products.

1.

17.11 The Friedel–Crafts Acylation 20260402

1. An **acyl group** is a carbonyl group with an alkyl group attached. Acyl groups are named systematically by dropping the final -e from the alkane name and adding the *-oyl* suffix.
2. An **acyl chloride** is an acyl group bonded to a chlorine atom. Acyl chlorides are made by reaction of the corresponding carboxylic acids with thionyl chloride. Therefore, acyl chlorides are also called **acid chlorides**.
3. In the presence of aluminum chloride, an acyl chloride reacts with benzene (or an activated benzene derivative) to give a phenyl ketone: an *acylbenzene*.
4. **Friedel–Crafts acylation** is analogous to the Friedel–Crafts alkylation, except that the reagent is an acyl chloride instead of an alkyl halide and the product is an acylbenzene (a "phenone") instead of an alkylbenzene.

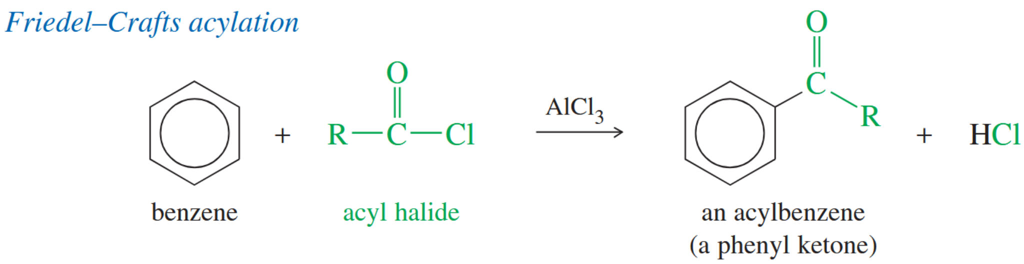


Figure 17.7 Friedel–Crafts acylation

17.11.1 Mechanism of Acylation

1. The mechanism of Friedel–Crafts acylation (shown next) resembles that for alkylation, except that the electrophile is a resonance-stabilized acylium ion. The acylium ion reacts with benzene or an activated benzene derivative via an electrophilic aromatic substitution to form an acylbenzene.

Mechanism 17.6 Friedel–Crafts Acylation

Friedel–Crafts acylation is an electrophilic aromatic substitution with an acylium ion acting as the electrophile.

2. Comparison of Friedel–Crafts Alkylation and Acylation

17.11.2 The Clemmensen Reduction: Synthesis of Alkylbenzenes

1. We use the Friedel–Crafts acylation to make the acylbenzene, then we reduce the acylbenzene to the alkylbenzene using the **Clemmensen reduction**: treatment with aqueous HCl and amalgamated zinc (zinc treated with mercury salts).
2. The reagents and conditions for the Clemmensen reduction are similar to those used to reduce a nitro group to an amine.

17.11.3 The Gatterman–Koch Formylation: Synthesis of Benzaldehydes

1. We cannot add a formyl group to benzene by Friedel–Crafts acylation in the usual manner. Formyl chloride is unstable and cannot be bought or stored.
2. Formylation can be accomplished by using a high-pressure mixture of carbon monoxide and HCl together with a catalyst consisting of a mixture of cuprous chloride (CuCl) and aluminum chloride.
3. The **Gatterman–Koch synthesis** is widely used in industry to synthesize aryl aldehydes.

17.12 Nucleophilic Aromatic Substitution 20260402

1. A nucleophile substitutes for a leaving group on an aromatic ring, and this class of reactions is called nucleophilic aromatic substitution.
2. Electron-withdrawing substituents (such as nitro groups) *activate* the ring toward nucleophilic aromatic substitution.

17.12.1 The Addition–Elimination Mechanism

1. Consider the reaction of 2,4-dinitrochlorobenzene with sodium hydroxide (shown next)...

Mechanism 17.7 Nucleophilic Aromatic Substitution (Addition–Elimination)

2. The resonance forms shown in the mechanism box illustrate how nitro groups ortho and para to the halogen help to stabilize the intermediate (and the transition state leading to it).

17.12.2 The Benzyne Mechanism: Elimination–Addition

1. Elimination–addition mechanism, called the **benzyne mechanism** because of the unusual intermediate.
2. Sodium amide (or sodium hydroxide in the Dow process) reacts as a base, abstracting a proton.

Mechanism 17.8 Nucleophilic Aromatic Substitution (Benzyne Mechanism) The benzyne mechanism (elimination–addition) is likely when the ring has no strong electron-withdrawing groups. It usually requires a powerful base or high temperatures.

- Step 1: Deprotonation adjacent to the leaving group gives a carbanion.
- Step 2: The carbanion expels the leaving group to give a "benzyne" intermediate.
- Step 3: The nucleophile attacks at either end of the reactive benzyne triple bond.
- Step 4: Reprotonation gives the product.

17.13 Aromatic Substitutions Using Organometallic Reagents 20260407

17.14 Addition Reactions of Benzene Derivatives 20260407

17.15 Side-Chain Reactions of Benzene Derivatives 20260407

17.15.1 Permanganate Oxidation

17.15.2 Side-Chain Halogenation

1. Alkylbenzenes undergo free-radical halogenation much more easily than alkanes because abstraction of a hydrogen atom at a benzylic position gives a resonancestabilized benzylic radical.

17.15.3 Nucleophilic Substitution at the Benzylic Position

1. Fisrt-Order Reactions: First-order nucleophilic substitution requires ionization of the halide to give a carbocation. In the case of a benzylic halide, the carbocation is resonance-stabilized.
2. Second-Order Reactions:

17.16 Reactions of Phenols 20260407

17.16.1 Oxidation of Phenols to Quinones

1. Phenols undergo oxidation, but they give different types of products from those seen with aliphatic alcohols. Chromic acid oxidation of a phenol gives a conjugated 1,4-diketone called a quinone.

17.16.2 Electrophilic Aromatic Substitution of Phenols

1. Phenols are highly reactive substrates for electrophilic aromatic substitution because the nonbonding electrons of the hydroxyl group stabilize the sigma complex formed by attack at the ortho or para position.
2. the hydroxyl group is strongly activating and ortho, para-directing. Phenols are excellent substrates for halogenation, nitration, sulfonation, and some Friedel–Crafts reactions.

Example 17.11 1,4-Benzoquinone is a good Diels–Alder dienophile. Predict the products of its reaction with

Example 17.12 Predict the major products of bromination of the following compounds, using and FeBr3 in the dark.

(a)

Example 17.13 Give the structures of compounds A through H in the following series of reactions.

Example 17.14 A student added 3-phenylpropanoic acid to a molten salt consisting of a 1:1 mixture of NaCl and $AlCl_3$ maintained at 170 °C. After 5 minutes, he poured the molten mixture into water and extracted it into dichloromethane. Evaporation of the dichloromethane gave a 96% yield of the product whose spectra follow. The mass spectrum of the product shows a molecular ion at m/z 132. What is the product?

SOLUTION indanone

Example 17.15 The following compound reacts with a hot, concentrated solution of NaOH (in a sealed tube) to give a mixture of two products. Propose structures for these products, and give a mechanism to account for their formation.

SOLUTION Considering benzyne

Example 17.16 *Bisphenol* A is an important component of many polymers, including polycarbonates, polyurethanes, and epoxy resins. It is synthesized from phenol and acetone with HCl as a catalyst. Propose a mechanism for this reaction.

Example 17.17 When anthracene is added to the reaction of chlorobenzene with concentrated NaOH at 350 °C, an interesting Diels–Alder adduct of formula $C_{20}H_{14}$ results. The proton NMR spectrum of the product shows a singlet of area 2 around δ 3 and a broad singlet of area 12 around δ 7. Propose a structure for the product, and explain why one of the aromatic rings of anthracene reacted as a diene.

SOLUTION NaOH and 350 °C, considering benzyne

Example 17.18 Phenol reacts with three equivalents of bromine in (in the dark) to give a product of formula When this product is added to bromine water, a yellow solid of molecular formula precipitates out of the solution. The IR spectrum of the yellow precipitate shows a strong absorption (much like that of a quinone) around Propose structures for the two products.

Example 17.19 A graduate student tried to make *o*-fluorophenylmagnesium bromide by adding magnesium to an ether solution of *o*-fluorobromobenzene. After obtaining puzzling results with this reaction, she repeated the reaction by using as solvent some tetrahydrofuran that contained a small amount of furan. From this reaction, she isolated a fair yield of the compound that follows. Propose a mechanism for its formation.

Example 17.20 The antioxidants BHA and BHT are commonly used as food preservatives. Show how BHA and BHT can be made from phenol and hydroquinone.

18 Ketones and Aldehydes

18.1 Carbonyl Compounds 20260409

1. carbonyl group (C═O)
2. Common Classes of Carbonyl Compounds
3. Ketone: Two alkyl groups bonded to a carbonyl group.
   Aldehyde: One alkyl group and one hydrogen bonded to a carbonyl group.

18.2 Structure of the Carbonyl Group 20260409

1. The carbonyl carbon atom is sp^2 hybridized and bonded to three other atoms.
2. The double bond of the carbonyl group has a large dipole moment because oxygen is more electronegative than carbon, and the bonding electrons are not shared equally.
3. Resonance forms. The first resonance form is more important.

18.3 Nomenclature of Ketones and AldehydesNomenclature of Ketones and Aldehydes 20260409

18.3.1 IUPAC Names

1. Ketone
   (1) Number the longest chain that includes the carbonyl carbon from the end closest to the carbonyl group.
   (2) Replace the final *-e* in the alkane name with *-one*.
2. Aldehydes
   (1) An aldehyde carbon is number 1.
   (2) Replace the final *-e* of the alkane name with *-al*.
   (3) If the aldehyde group is a substituent of a large unit (usually a ring), the suffix *carbaldehyde* is used.
   cyclohexanecarbaldehyde, 2-hydroxycyclopentane-1-carbaldehyde
3. A ketone or aldehyde group can also be named as a substituent on a molecule with a higher priority functional group as its root.
   (1) A ketone or aldehyde carbonyl is named by the prefix *oxo* if it is included as part of the longest chain in the root name.
   (2) When an aldehyde group is a substituent and not part of the longest chain, it is named

by the prefix formyl.

18.3.2 Common Names

1. Ketone common names are formed by naming the two alkyl groups bonded to the carbonyl group.
2. Some ketones have historical common names.
3. Common names of aldehydes are derived from the common names of the corresponding carboxylic acids. Greek letters are used with common names of aldehydes to give the locations of substituents.

18.4 Physical Properties of Ketones and Aldehydes 20260409

1. Polarization of the carbonyl group creates dipole–dipole attractions between the molecules of ketones and aldehydes, resulting in higher boiling points than for hydrocarbons and ethers of similar molecular weights.
2. Ketones and aldehydes have no O — H or N — H bonds, however, so their molecules cannot form hydrogen bonds with each other. Their boiling points are therefore lower than those of alcohols of similar molecular weight.
3. H bond: Traditional (D-H···A), C-H···Cl, M-H, C-H···Benzene.
4. The higher the *s* orbital ratio, the stronger the base.
5. Ketones and aldehydes have lone pairs of electrons and can act as hydrogen bond acceptors with other compounds having or O — H or N — H bonds. Ketones and aldehydes are good solvents for polar hydroxylic substances such as alcohols.
6. Formaldehyde and acetaldehyde are the most common aldehydes.
7. Formaldehyde is a gas at room temperature, so it is often stored and used as a 40% aqueous solution called *formalin*.
8. When dry formaldehyde is needed, it can be generated by heating one of its solid derivatives. Trioxane is a cyclic trimer, containing three formaldehyde units. Paraformaldehyde is a linear polymer, containing many formaldehyde units.
9. Acetaldehyde boils near room temperature.
10. Acetaldehyde is also used as a trimer (paraldehyde) and a tetramer (metaldehyde).

18.5 Spectroscopy of Ketones and Aldehydes 20260409

18.5.1 Infrared Spectra of Ketones and Aldehydes

1. The carbonyl stretching vibrations of simple ketones occur around $1710\,\mathrm{cm}^{-1}$, and simple aldehydes around $1725\,\mathrm{cm}^{-1}$.
2. In addition to the carbonyl absorption, an aldehyde shows a set of two low-frequency stretching absorptions around 2710 and 2810 cm -1.

18.5.2 Proton NMR Spectra of Ketones and Aldehydes

1. Aldehyde protons normally appear at chemical shifts between δ 9 and δ 10.
2. The aldehyde proton's absorption may be split ($J = 1$ to 5 Hz) if there are protons on the atom.

18.5.3 Carbon NMR Spectra of Ketones and Aldehydes

1. The carbonyl carbon atoms of aldehydes and ketones have chemical shifts around 200 ppm in the carbon NMR spectrum.

18.5.4 Mass Spectra of Ketones and Aldehydes

1. α cleavage
2. McLafferty Rearrangement of Ketones and Aldehydes: a cyclic intramolecular transfer of a hydrogen atom from the γ carbon to the carbonyl oxygen.
3. γ hydrogen is needed.
4. The McLafferty rearrangement is a characteristic fragmentation of ketones and aldehydes as long as they have hydrogens. It is equivalent to a cleavage between the α and $beta$ carbon atoms, plus one mass unit for the hydrogen that is transferred.
5. The fragment from the McLafferty rearrangement has an even-numbered mass.
6. enol

18.5.5 Ultraviolet Spectra of Ketones and Aldehydes

1. The $\pi \to \pi^*$ Transition

18.6 Industrial Importance of Ketones and Aldehydes 20260409

18.7 Review of Syntheses of Ketones and Aldehydes 20260414

18.7.1 Ketones and Aldehydes from Oxidation of Alcohols

1. Often use a Grignard reagent to synthesize an alcohol with the correct structure and oxidize it to the final product.
   (1) Secondary alcohols $\rightarrow$ ketones: Sodium dichromate in sulfuric acid ("chromic acid," H_2CrO_4) is the traditional laboratory reagent for oxidizing secondary alcohols to ketones.
   (2) Primary alcohols $\rightarrow$ aldehydes: Pyridinium chlorochromate (PCC), a complex of chromium trioxide with pyridine and HCl, provides good yields of aldehydes without over-oxidation.

18.7.2 Ketones and Aldehydes from Ozonolysis of Alkenes

1. Ozonolysis, followed by a mild reduction, cleaves alkenes to give ketones and aldehydes.
2. Ozonolysis is useful as a synthetic method.

18.7.3 Phenyl Ketones and Aldehydes: Friedel–Crafts Acylation

1. Friedel–Crafts acylation is an excellent method for making alkyl aryl ketones or diaryl ketones.

18.7.4 Ketones and Aldehydes from Hydration of Alkynes

1. Catalyzed by Acid and Mercuric Salts
   (1) Hydration of a terminal alkyne, catalyzed by a combination of sulfuric acid and mercuric ion.
   (2) The initial product of Markovnikov hydration is an enol.
2. Hydroboration–Oxidation of Alkynes
   (1) Hydroboration–oxidation of an alkyne gives antiMarkovnikov addition of water across the triple bond.
   (2) On oxidation of the borane, the unstable enol quickly tautomerizes to an aldehyde.

18.8 Synthesis of Ketones and Aldehydes Using 1,3-Dithianes 20260414

1. 1,3-Dithiane can be deprotonated by strong bases such as *n*-butyllithium.

2. Alkylation of the dithiane anion by a primary alkyl halide or a tosylate gives a **thioacetal** that can be hydrolyzed into the aldehyde by using **an acidic solution of mercuric chloride**.
3. The thioacetal can be isolated and deprotonated. Alkylation gives a **thioketal**, then hydrolysis will produce a ketone.

18.9 Synthesis of Ketones from Carboxylic Acids 20260414

1. Organolithium reagents can be used to synthesize ketones from carboxylic acids.
2. Organolithiums will attack the lithium salts of carboxylate anions to give dianions.
3. Protonation of the dianion forms the hydrate of a ketone, which quickly loses water to give the ketone.

18.10 Synthesis of Ketones and Aldehydes from Nitriles 20240414

1. Nitriles are compounds containing the cyano ($-C \equiv N$) functional group.
2. A Grignard or organolithium reagent can attack the carbon of the nitrile.
3. The imine is then hydrolyzed to form a ketone.
4. Aluminum hydrides can reduce nitriles to the corresponding aldehydes. Diisobutylaluminum hydride, abbreviated (i-Bu)2AlH or **DIBAL-H**, is commonly used for the reduction of nitriles (mild reducing agent).

18.11 Synthesis of Aldehydes and Ketones from Acid Chlorides and Esters 20260414

1. Acids can be reduced to aldehydes by first converting them to acid chloride.
2. Lithium tri-*tert*-butoxyaluminum hydride is a milder reducing agent that reacts faster with acid chlorides than with aldehydes.
3. Synthesis of Ketones
   (1) Grignard and organolithium reagents add to acid chlorides to give ketones, but they add again to the ketones to give tertiary alcohols.
   (2) **Lithium dialkylcuprates** (Gilman reagent) and other organocuprates react with acid chlorides to give good yields of a wide variety of ketones.
   (3) The lithium dialkylcuprate is formed by the reaction of two equivalents of the corresponding organolithium reagent with cuprous iodide.
4. Syntheses of Ketones and Aldehydes

18.12 Reactions of Ketones and Aldehydes: Introduction to Nucleophilic Addition 20260414

1. Strong Nu: R^-, H^-, $NaNH_2$, RO^-, NC^-, $C{\equiv}C^-$
   Weak Nu: ROH, RNH_2, H_2O.
2. Weak nucleophiles can add to activated carbonyl group under acidic conditions. A carbonyl group that is protonated (or bonded to some other electrophile) is strongly electrophilic, inviting attack by a weak nucleophile.
3. Aldehydes are more reactive than ketones toward nucleophilic additions. The enhanced reactivity of aldehydes is due to an electronic effect and a steric effect.

Example 18.1 Sodium triacetoxyborohydride, $NaBH(OAc)_3$, is a mild reducing agent that reduces aldehydes much more quickly than ketones. It can be used to reduce aldehydes in the presence of ketones, such as in the following reaction:

Mechanism 18.1 Nucleophilic Additions to Carbonyl Groups

1. Basic Conditions (strong nucleophile)
   - Step 1: A strong nucleophile adds to the carbonyl group to form an alkoxide.
   - Step 2: A weak acid protonates the alkoxide to give the addition product.
2. Acidic Conditions (weak nucleophile, activated carbonyl)
   hemiacetal

18.13 The Wittig Reaction 20260414 20260416

1. Adding a phosphorus-stabilized carbanion to a ketone or aldehyde, the Wittig reaction converts the carbonyl group into a new $C{=}C$ double bond where no bond existed before.
2. A phosphorus ylide is used as the nucleophile in the reaction.
3. Triphenylphosphine oxide is exceptionally stable, and the conversion of triphenylphosphine to triphenylphosphine oxide provides the driving force for the Wittig reaction.
4. Mixtures of cis and trans isomers often result.

Mechanism 18.2 The Wittig Reaction

- Step 1: The ylide attacks the carbonyl to form a betaine.
- Step 2: The betaine closes to a four-membered ring oxaphosphetane (first $P{-}O$ bond formed).
- Step 3: The ring collapses to the products (second $P{-}O$ bond formed).

Example 18.2 Like other strong nucleophiles, triphenylphosphine attacks and opens epoxides.

The initial product (a betaine) quickly cyclizes to an oxaphosphetane that collapses to an alkene and triphenylphosphine oxide.

(a) Show each step in the reaction of *trans*-2,3-epoxybutane with triphenylphosphine to give but-2-ene. What is the stereochemistry of the double bond in the product?
(b) Show how this sequence might be used to convert *cis*-cyclooctene to *trans*-cyclooctene.

5. Planning a Wittig Synthesis
   (1) In general, the ylide should come from an unhindered alkyl halide.

Example 18.3 Show how Wittig reactions might be used to synthesize the following compounds. In each case, start with an alkyl halide and a ketone or an aldehyde.

(a) $\mathrm{Ph{-}CH{=}CH_3}$
(b) $\mathrm{Ph{-}CH_3{=}CH_2}$

18.14 Hydration of Ketones and Aldehydes 20260416

1. In an aqueous solution, a ketone or an aldehyde is in equilibrium with its hydrate, a geminal diol.
2. With most ketones, the equilibrium favors the unhydrated keto form of the carbonyl.
3. Aldehydes are more likely than ketones to form stable hydrates, because an aldehyde carbonyl has only one stabilizing alkyl group. The partial positive charge of the aldehyde is not as well stabilized.

Mechanism 18.3 Hydration of Ketones and Aldehydes

1. **In acid**
   The acid-catalyzed hydration is a typical acid-catalyzed addition to the carbonyl group. Protonation, followed by addition of water, gives a protonated product. Deprotonation gives the hydrate.
   - Step 1: Protonation.
   - Step 2: Water adds.
   - Step 3: Deprotonation.

18.15 Formation of Cyanohydrins 20260416

1. Cyanide ion is a strong base and a strong nucleophile. It attacks ketones and aldehydes to give addition products called cyanohydrins.

2. These equilibrium constants follow the general reactivity trend of ketones and aldehydes:

$$\text{formaldehyde} > \text{other aldehydes} > \text{ketones}$$

Mechanism 18.4 Formation of Cyanohydrins

Cyanohydrin formation is a perfect example of base-catalyzed addition to a carbonyl group. The strong nucleophile adds in the first step to give an alkoxide. Protonation gives the cyanohydrin.

- Step 1: Cyanide adds to the carbonyl.
- Step 2: Protonation gives the cyanohydrin.

3. Organic compounds containing the cyano group are called nitriles. A cyanohydrin is therefore an α-hydroxynitrile.
4. Nitriles hydrolyze to carboxylic acids under acidic conditions, so cyanohydrins hydrolyze to α-hydroxy acids.

18.16 Formation of Imines 20260416

1. Ammonia or a primary amine reacts with a ketone or an aldehyde to form an imine.
2. Imines are nitrogen analogues of ketones and aldehydes, with a $\mathrm{C{=}N}$ in place of the carbonyl group.
3. Imines are basic, a substituted imine is also called a **Schiff base**.
4. The proper pH is crucial to imine formation, fastest around pH 4.5.

Mechanism 18.5 Formation of Imines

1. First part: Acid-catalyzed addition of the amine to the carbonyl group.
   - Step 1: Protonation of the carbonyl.
   - Step 2: Addition of the amine.
   - Step 3: Deprotonation.
2. Second part: Acid-catalyzed dehydration.
   - Step 4: Protonation of the $-\mathrm{OH}$ group.
   - Step 5: Loss of H_2O.
   - Step 6: Deprotonation.

5. Imine formation is reversible, and most imines can be hydrolyzed back to the amine and the ketone or aldehyde.
6. Enamine Formation

   An enamine results from the reaction of a ketone or aldehyde with a secondary amine. Activate the α place of carbonyl group.

18.17 Condensations with Hydroxylamine and Hydrazines 20260416

1. Condensations of Amines with Ketones and Aldehydes
2. oxime hydrazine hydrazone phenylhydrazine phenylhydrazone semicarbazide semicarbazone

Example 18.4 2,4-Dinitrophenylhydrazine is frequently used for making derivatives of ketones and aldehydes because the products (2,4-dinitrophenylhydrazones, called 2,4-DNP derivatives) are even more likely than the phenylhydrazones to be solids with sharp melting points. Propose a mechanism for the reaction of acetone with 2,4-dinitrophenylhydrazine in a mildly acidic solution.

It is used in test for glucose.

Example 18.5 Predict the products of the following reactions.

(a)

18.18 Formation of Acetals 20260416

1. Ketones and aldehydes react with alcohols to form acetals.
2. In the formation of an acetal, two molecules of alcohol add to the carbonyl group, and one molecule of water is eliminated.
3. Acetal formation must be acidcatalyzed.

Mechanism 18.6 Formation of Acetals

1. First half: Acid-catalyzed addition of the alcohol to the carbonyl group.
   - Step 1: Protonation.
   - Step 2: Alcohol adds.
   - Step 3: Deprotonation.
2. Second half: $\mathrm{S_N1}$ substitution of the protonated hemiacetal.
   - Step 4: Protonation.
   - Step 5: Loss of water.

4. Equilibrium of Acetal Formation
5. Cyclic Acetala
   (1) Formation of an acetal using a diol as the alcohol gives a cyclic acetal.
   (2) Ethylene glycol is often used to make cyclic acetals; its acetals are called ethylene acetals (or ethylene ketals).
   (3) The reaction is reversible and is used in synthesis to protect carbonyls from reaction

18.19 Use of Acetals as Protecting Groups 20260416 20260421

Example 18.6 Show how you would accomplish the following syntheses. You may use whatever additional reagents you need.

18.20 Oxidation of Aldehydes 202640416

1. Aldehydes are easily oxidized to carboxylic acids by common oxidants such as bleach (sodium hypochlorite), chromic acid, permanganate, and peroxy acids.
2. Silver ion oxidizes aldehydes selectively in a convenient functional-group test for aldehydes.
3. The **Tollens test** involves adding a solution of silver–ammonia complex (the **Tollens reagent**) to the unknown compound.

Example 18.7 f

18.21 Reductions of Ketones and Aldehydes 20250421

18.21.1 Hydride Reductions (Review)

18.21.2 Catalytic Hydrogenation

18.21.3 Deoxygenation of Ketones and Aldehydes

1. Clemmensen Reduction (Review)
2. Wolff–Kishner Reduction: The ketone or aldehyde is converted to its hydrazone, which is heated with a strong base such as KOH or potassium tert-butoxide. Compounds that cannot survive treatment with hot acid can be deoxygenated using the Wolff–Kishner reduction.

Assignment 18-39, 48, 50/abcd, 56, 60, 61

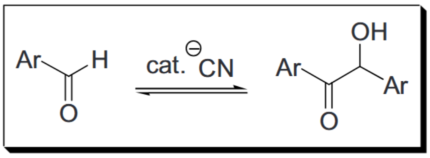
Example 18.8 Rank the following carbonyl compounds in order of increasing equilibrium constant for hydration:

CH_3COCH_3Cl $\quad ClCH_2CHO$ $\quad CH_2O$ $\quad CH_3COCH_3$ $\quad CH_3CHO$

Example 18.9 The proton NMR spectrum of a compound of formula $C_{10}H_{12}O$ follows. This compound reacts with an acidic solution of 2,4-dinitrophenylhydrazine to give a crystalline derivative, but it gives a negative Tollens test. Propose a structure for this compound and give peak assignments to account for the signals in the spectrum.

Example 18.10 Solving the following road-map problem depends on determining the structure of **A**, the key intermediate. Give structures for compounds **A** through **K**.

1. Benzoin Condensation



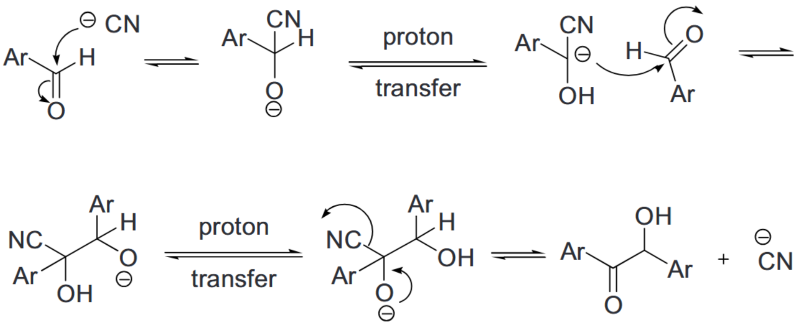


Figure 18.1 Benzoin Condensation

19 Amines

19.1 Introduction 20260421

1. Amines are organic derivatives of ammonia with one or more alkyl or aryl groups bonded to the nitrogen atom.
2. dopamine, epinephrine, L-tryptophan, piperazine,
3. *Alkaloids*

19.2 Nomenclature of Amines 20260421 20260423

1. Amines are classified as **primary** (1°), **secondary** (2°), or **tertiary** (3°), quaternary (4°) corresponding to one, two, three or four alkyl or aryl groups bonded to nitrogen.
2. **Quaternary ammonium salts** have four alkyl or aryl bonds to a nitrogen atom.

19.2.1 Common Names

1. Common names of amines are formed from the names of the alkyl groups bonded to nitrogen, followed by the suffix *-amine*.
2. In naming amines with more complicated structures, the $-NH_2$ group is called the amino group. Secondary and tertiary amines are named by classifying the nitrogen atom (together with its alkyl groups) as an alkylamino group.
3. Phenylamine is called *aniline*, for example, and its derivatives are named as derivatives of aniline.
4. Nitrogen heterocycles. The heteroatom is usually assigned position number 1. aziridine, pyrrole, pyrrolidine, imidazole, indole, pyridine, piperidine, pyrimidine, purine

19.2.2 IUPAC Names

1. The IUPAC nomenclature for amines is similar to that for alcohols.
2. The longest continuous chain of carbon atoms determines the root name. The *-e* ending in the alkane name is changed to *-amine*, and a number shows the position of the amino group along the chain.
3. butan-2-amine, *N*-methylbutan-2-amine, 2,4,*N*,*N*-tetramethylhexan-3-amine

19.3 Structure of Amines 20260423

1. Ammonia has a slightly distorted tetrahedral shape.
2. A tetrahedral amine with three different substituents (and a lone pair) is nonsuperimposable on its mirror image, and appears to be a chirality center.
3. Cannot be resolved two enantiomers because the enantiomers interconvert rapidly, **nitrogen inversion**.

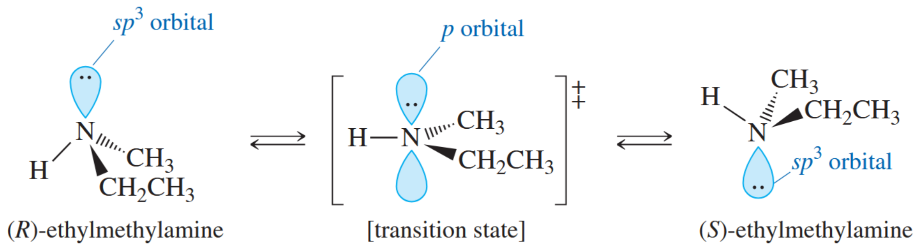


Figure 19.1 Nitrogen Inversion

4. Several types of chiral amines can be resolved:
   (1) Amines whose chirality stems from the presence of asymmetric carbon atoms.
   (2) Quaternary ammonium salts with asymmetric nitrogen atoms(larger than hydrogen).
   (3) Amines that cannot attain the sp^2 transition state for nitrogen inversion.

Example 19.1 Which of the amines listed next can be resolved into enantiomers? In each case, explain why interconversion of the enantiomers would or would not take place.

(a) *cis*-2-methylcyclohexanamine
(b) *N*-ethyl-*N*-methylcyclohexanamine
(c) *N*-methylaziridine
(d) ethylmethylanilinium iodide
(e) methylethylpropylisopropylammonium iodide

Example 19.2 Which of the following compounds are capable of being resolved into enantiomers?

19.4 Physical Properties of Amines 20260423

1. Nitrogen is less electronegative than oxygen. Amines form weaker hydrogen bonds than do alcohols of similar molecular weights.
2. Amines tend to be soluble in alcohols, and the lowermolecular-weight amines (up to about four carbon atoms) are relatively soluble in water.

3. Perhaps the most obvious property of amines is their characteristic odor of rotting fish. putrescine cadaverine

19.5 Basicity of Amines 20260423

1. An amine is a nucleophile (a Lewis base) because its lone pair of nonbonding electrons can form a bond with an electrophile. An amine can also act as a Brønsted–Lowry base by accepting a proton from a proton acid.
2. **Base-dissociation constant** for the amine, symbolized by $\mathbf{K}_\mathrm{b}$.

19.6 Effects on Amine Basicity 20260423 20260428

1. Any structural feature that stabilizes the ammonium ion (relative to the free amine) shifts the reaction toward the right, making the amine a stronger base.
2. Substitution by Alkyl Groups
   (1) Alkyl groups are electron-donating toward cations.
   (2) The simple alkylamines are generally stronger bases than ammonia.
3. Resonance Effects on Basicity
   (1) Arylamines (anilines ($\mathrm{p}K_\mathrm{b} = 9.40$) and their derivatives) are much weaker bases than simple aliphatic amines. This reduced basicity is due to resonance delocalization of the nonbonding electrons in the free amine.
   (2) Resonance effects also influence the basicity of pyrrole. When the pyrrole nitrogen is protonated, pyrrole loses its aromatic stabilization.
4. Hybridization Effects
   (1) Electrons are held more tightly by orbitals with more s character.
   (2) Pyridine's nonbonding electrons are less available for bonding to a proton. ($\mathrm{p}K_\mathrm{b} = 8.75$)

Example 19.3 Rank each set of compounds in order of increasing basicity.

(a) NaOH, NH_3, CH_3NH_2, $Ph-NH_2$

Example 19.4 Rank the amines in each set in order of increasing basicity.

Example 19.5 Complete the following proposed acid–base reactions, and predict whether the reactants or products are favored.

19.7 Salts of Amines 2060428

1. Protonation of an amine gives an amine salt.
2. Simple amine salts are named as the substituted **ammonium salts**. Salts of complex amines use the names of the amine and the acid that make up the salt.
3. Amine salts are ionic, high-melting, nonvolatile solids, much more soluble in water than the parent amines, and only slightly soluble in nonpolar organic solvents.
4. Formation of amine salts can be used to isolate and characterize amines.
5. We can use the formation of amine salts to separate amines from less basic compounds.
6. Many drugs and other biologically important amines are commonly stored and used as their salts.

19.8 Phase Transfer Catalysts 20260428

19.9 Spectroscopy of Amines

19.9.1 Infrared Spectroscopy

1. The most reliable IR absorption of primary and secondary amines is the $\mathrm{N{-}H}$ stretch whose frequency appears between 3200 and $3500\,\mathrm{cm}^{-1}$.

19.9.2 Proton NMR Spectroscopy

19.9.3 Carbon NMR Spectroscopy

19.9.4 Mass Spectrometry

1. Nitrogen has an odd valence and an even mass number.
2. Whenever an odd number of nitrogen atoms are present in a molecule, the molecular ion has an odd mass number. Most of the fragments have even mass numbers.

19.10 Reactions of Amines with Ketones and Aldehydes (Review) 20260428

1.

19.11 Aromatic Substitution of Arylamines and Pyridine 20260428

19.11.1 Electrophilic Aromatic Substitution of Arylamines

1. The $-NH_3^+$ group is strongly deactivating (and meta-allowing). Therefore, strongly acidic reagents are unsuitable for substitution of anilines.

19.11.2 Electrophilic Aromatic Substitution of Pyridine

1. Pyridine resembles nitro benzene in its reactivity toward electrophilic aromatic substitution.
2. Friedel–Crafts reactions fail completely, and other substitutions require unusually strong conditions.
3. Deactivation results from the electron-withdrawing effect of the electronegative nitrogen atom.

19.11.3 Nucleophilic Aromatic Substitution of Pyridine

19.12 Alkylation of Amines by Alkyl Halides 20260428

1. Amines react with primary alkyl halides to give alkylated ammonium halides.
2. Alkylation proceeds by the S_N2 mechanism, so it is not feasible with tertiary halides because they are too hindered.
3. Even if just one equivalent of the halide is added, some amine molecules will react once, some will react twice, and some will react three times.

19.13 Acylation of Amines by Acid Chlorides 20260428

1. Primary and secondary amines react with acid halides to form amides.
2. This reaction is a *nucleophilic acyl substitution*: the replacement of a leaving group on a carbonyl carbon by a nucleophile.

Mechanism 19.1 Acylation of an Amine by an Acid Chloride

- Step 1: A nucleophile attacks the strongly electrophilic carbonyl group of the acid chloride to form a tetrahedral intermediate.
- Step 2: The tetrahedral intermediate expels chloride ion.
- Step 3: Loss of a proton gives the amide.

3. The diminished basicity of amides can be used to advantage in **electrophilic aromatic substitutions**.

4. If the amino group of aniline is acetylated to give acetanilide, the resulting amide is still activating and ortho, para-directing.
5. Aryl amino groups are frequently acylated before further substitutions are attempted on the ring, and the acyl group is removed later by acidic or basic hydrolysis

Example 19.6 Show how you would accomplish the following synthetic conversion in good yield.

SOLUTION An attempted Friedel–Crafts acylation on aniline would likely meet with disaster. The free amino group would attack both the acid chloride and the Lewis acid catalyst.

19.14 Formation of Sulfonamides 20264028

1.

19.15 Amines as Leaving Groups: The Hofmann Elimination 20260428

1. An amine cannot undergo elimination directly because the leaving group would be an amide ion ($^{-}NH_2$ or ^{-}NHR), which is a very strong base and a poor leaving group.
2. An amino group can be converted to a good leaving group by exhaustive methylation, which converts it to a quaternary ammonium salt that can leave as a neutral amine.
3. Exhaustive methylation is usually accomplished using methyl iodide.
4. Elimination of the quaternary ammonium salt generally takes place by the E2 mechanism, which requires a strong base. To provide the base, the quaternary ammonium iodide is converted to the hydroxide salt by treatment with silver oxide.
5. In the Hofmann elimination the product is commonly the *least-substituted* alkene (*Hofmann product*).
6. Remember that the E2 mechanism requires an **anti-coplanar arrangement** of the proton and the leaving group.
7. The Hofmann product predominates because elimination of one of the C1 protons involves a lower-energy, more probable transition state than the crowded transition state required for Zaitsev ($C2-C3$) elimination.
8. The least-substituted alkene is usually the favored product.

Example 19.7 Predict the major product(s) formed when the following amine is treated with excess iodomethane, followed by heating with silver oxide.

Example 19.8 Predict the major products formed when the following amines undergo exhaustive methylation, treatment with and heating.

(a) hexan-2-amine
(b) 2-methylpiperidine
(c) *N*-ethylpiperidine

19.16 Oxidation of Amines; The Cope Elimination 20260430

1. sdf

Example 19.9 MCPBA

19.17 Reactions of Arenediazonium Salts 20260430 20260507

1. d
2. These reactions of diazonium salts are **extremely useful** for solving aromatic synthesis problems.

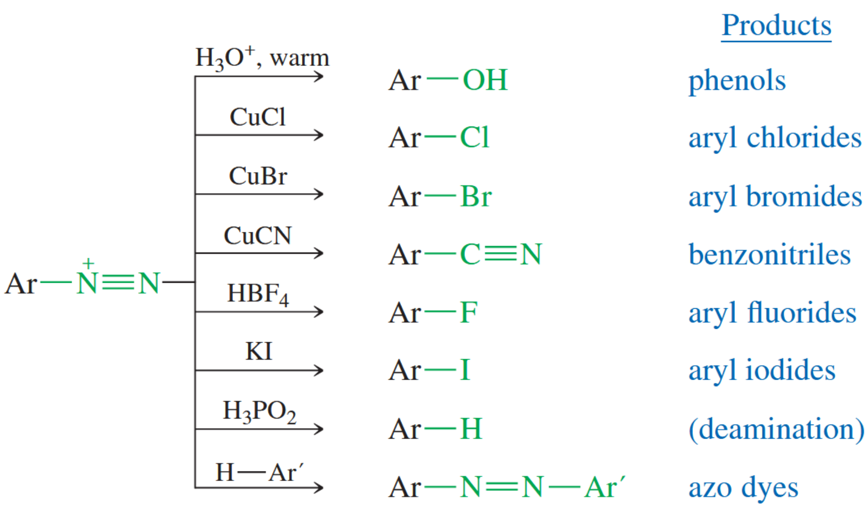


Figure 19.2 Reactions of Diazonium Salts

3. Replacement of the Diazonium Group by Hydroxide: Hydrolysis
   (1) Hydrolysis takes place when the acidic solution of an arenediazonium salt is warmed.
   (2) This is a useful laboratory synthesis of phenols.
4. Replacement of the Diazonium Group by Chloride, Bromide, and Cyanide: The Sandmeyer Reaction
   (1) The use of cuprous salts to replace arenediazonium groups is called the Sandmeyer reaction.
   (2) The Sandmeyer reaction (using cuprous cyanide) is also an excellent method for attaching another carbon substituent to an aromatic ring.

5. Replacement of the Diazonium Group by Fluoride and Iodide
   (1) When an arenediazonium salt is treated with fluoroboric acid HBF_4, the diazonium fluoroborate precipitates out of solution.
   (2) Aryl iodides are formed by treating arenediazonium salts with potassium iodide.
6. Reduction of the Diazonium Group to Hydrogen: Deamination of Anilines
7. Diazonium Salts as Electrophiles: Diazo Coupling
   (1) Arenediazonium ions act as weak electrophiles in electrophilic aromatic substitutions.
   (2) The product have the structure $\mathrm{Ar-N=N-Ar'}$.
8. Summary: Reactions of Amines

19.18 Synthesis of Amines by Reductive Amination 20260507

1. **Reductive amination** is the most general amine synthesis, capable of adding a primary or secondary alkyl group to an amine.
2. First we form an imine or oxime derivative of a ketone or aldehyde, and then reduce it to the amine.
3. Primary Amines
   (1) Primary amines result from condensation of hydroxylamine (zero alkyl groups) with a ketone or an aldehyde, followed by reduction of the oxime.
   (2) The oxime is reduced using catalytic reduction, lithium aluminum hydride, or zinc and HCl.
4. Secondary Amines
   (1) Condensation of a primary amine with a ketone or aldehyde forms an N-substituted imine (a Schiff base).
   (2) Reduction of the imine, using either $LiAlH_4$ or $NaBH_4$, gives a secondary amine.
5. Tertiary Amines
   (1) Condensation of a secondary amine with a ketone or aldehyde gives an iminium salt.
   (2) Iminium salts are frequently unstable, so they are rarely isolated.
   (3) Sodium triacetoxyborohydride [$NaBH(OCOCH_3)_3$ or $NaBH(OAc)_3$] is less reactive than sodium borohydride, and it does not reduce the carbonyl group.

19.19 Synthesis of Amines by Acylation–Reduction 20260507

1. The second general synthesis of amines is **acylation–reduction**.
2. Reduction of the amide by lithium aluminum hydride ($LiAlH_4$) gives the corresponding amine (no hydroxyl generated).
3. Primary amines

4. Secondary Amines
5. Tertiary Amines

19.20 Syntheses Limited to Primary Amines 20260507

19.20.1 Direct Alkylation and Gabriel Synthesis

1. Phthalimide
2. The phthalimide anion is a strong nucleophile, displacing a halide or tosylate ion from a good S_N2 substrate.
3. Heating the *N*-alkyl phthalimide with hydrazine displaces the primary amine, giving the very stable hydrazide of phthalimide.

19.20.2 Reduction of Azides and Nitriles

1. Formation and Reduction of Azides
2. Formation and Reduction of Nitriles
   (1) cyanide ion is a good nucleophile
   (2) Nitriles are reduced to primary amines by lithium aluminum hydride or by catalytic hydrogenation.

19.20.3 The Hofmann Rearrangement of Amides

1. In the presence of a strong base, primary amides react with chlorine or bromine to form shortened amines, with the loss of the carbonyl carbon atom.
2. This reaction, called the Hofmann rearrangement, is used to synthesize primary and aryl amines.
3. isocyanate, carbamic acid
4. The Hofmann rearrangement, can produce primary amines with primary, secondary, tertiary alkyl group or aryl amines.
5. The conformation will keep the same.

Example 19.10

Example 19.11 *Curtius rearrangement*

Assigment 19-36(no classification), 41, 42

Example 19.12 Ketones and aldehydes react with primary amines to give imines.

20 Carboxylic Acids

20.1 Introduction 20260512

1. The combination of a carbonyl group and a hydroxyl on the same carbon atom is called a **carboxyl group**.
2. An **aliphatic acid** has an alkyl group bonded to the carboxyl group, and **an aromatic** acid has an aryl group. **Fatty acids** are long-chain aliphatic acids derived from the hydrolysis of fats and oils.
3. A carboxylic acid donates protons by heterolytic cleavage of the acidic $\mathrm{O-H}$ bond to give a proton and a **carboxylate ion**.

20.2 Nomenclature of Carboxylic Acids 20260512

20.2.1 Commonn Names

1. Few common names:

20.2.2 IUPAC Names

1. The IUPAC nomenclature for carboxylic acids uses the name of the alkane/alkene that corresponds to the longest continuous chain of carbon atoms. The final *-e* in the alkane name is replaced by the suffix *-oic acid*.
2. Cycloalkanes with $-\mathrm{COOH}$ substituents are generally named as *cycloalkanecarboxylic acids*. e.g. 3,3-dimethylcyclohexanecarboxylic acid
3. Aromatic acids of the form $\mathrm{Ar-COOH}$ are named as derivatives of *benzoic acid*, $\mathrm{Ph-COOH}$.

20.2.3 Nomenclature of Dicarboxylic Acids

1. Common Names of Dicarboxylic Acids
   (1) A dicarboxylic acid (also called a *diacid*) is a compound with two **carboxyl groups**.
   (2) oxalic, malonic, maleic (can lose water), fumaric (cannot lose water), phthalic

1	2

2. IUPAC Names of Dicarboxylic Acids

(1) Aliphatic dicarboxylic acids are named simply by adding the suffix *-dioic acid* to the name of the parent alkane.

Example 20.1 Draw the structures of the following carboxylic acids.

Example 20.2 Name the following carboxylic acids (when possible, give both a common name and a systematic name).

20.3 Structure and Physical Properties of Carboxylic Acids 20260512

1. Structure of the Carboxyl Group
   (1) The entire molecule is approximately planar. The *sp*^{2} hybrid carbonyl carbon atom is planar, with nearly trigonal bond angles. The $\mathrm{O{-}H}$ bond also lies in this plane, eclipsed with the $\mathrm{C{=}O}$ bond.
   (2) One of the unshared electron pairs on the hydroxyl oxygen atom is delocalized into the electrophilic pi system of the carbonyl group.
2. Boiling Points
   (1) Carboxylic acids boil at considerably higher temperatures than do alcohols, ketones, or aldehydes of similar molecular weights.
   (2) The high boiling points of carboxylic acids result from formation of a stable, hydrogenbonded dimer.
3. Melting Points
4. Solubilities

20.4 Acidity of Carboxylic Acids 20260512

20.4.1 Measurement of Acidity

1. A carboxylic acid may dissociate in water to give a proton and a carboxylate ion. The equilibrium constant K_a for this reaction is called the *acid-dissociation constant*.
2. benzoic acid $\mathrm{p}K_\mathrm{a} = 4.19$ phenol $\mathrm{p}K_\mathrm{a} \sim 10$

20.4.2 Substituent Effects on Acidity

1. The magnitude of a substituent effect depends on its distance from the carboxyl group.
2. Substituted benzoic acids show similar trends in acidity, with electron-withdrawing groups enhancing the acid strength and electron-donating groups decreasing the acid strength. These effects are strongest for substituents in the ortho and para positions.

Example 20.3 Rank the compounds in each set in order of increasing acid strength.

(a) 1

(b) 2

(c) 3 $-NO_2$ group strongest

20.5 Salts of Carboxylic Acids 20260512

1. A strong base can completely deprotonate a carboxylic acid. The combination of a carboxylate ion and a cation is a **salt of a carboxylic acid**.
2. Nomenclature of Carboxylic Acid Salts
   (1) Salts of carboxylic acids are named simply by naming the cation, then naming the carboxylate ion by replacing the *-ic acid* part of the acid name with *-ate*.
3. Properties of Acid Salts

20.6 Commercial Sources of Carboxylic Acids 20260512

1. Hydrolysis of a fat or an oil gives a mixture of the salts of straight-chain fatty acids.
2. Important Acids
   (1) The most important commercial aliphatic acid is acetic acid.
   (2) Aromatic acids
   (3) Two important commercial diacids are adipic acid (hexanedioic acid) and terephthalic acid (benzene-1,4-dicarboxylic acid).

20.7

20.8

20.9

20.10 Condensation of Acids with Alcohols: The Fischer Esterification 20260514

1. The **Fischer esterification** converts carboxylic acids and alcohols directly to esters by an acid-catalyzed nucleophilic acyl substitutison.

20.11 Esterification Using Diazomethane 20260519

1. Carboxylic acids are converted to their methyl esters very simply by adding an ether solution of diazomethane.
2. Diazomethane is a toxic, explosive yellow gas that dissolves in ether and is fairly safe to use in ether solutions.

Mechanism 20.1 Esterification Using Diazomethane

20.12 Condensation of Acids with Amines: Direct Synthesis of Amides 20260519

1. The initial acid–base reaction of a carboxylic acid with an amine gives an ammonium carboxylate salt.
2. Heating this salt to well above $100\,^{\circ}\mathrm{C}$ drives off steam and forms an amide.

20.13 Reduction of Carboxylic Acids 20260519

1. Lithium aluminum hydride ($LiAlH_4$ or LAH) reduces carboxylic acids to primary alcohols.
2. Possible mechanism
3. Borane also reduces carboxylic acids to primary alcohols.
4. React with the carboxyl group faster than with any other carbonyl function, elcellent selectivity.
5. Reduction to Aldehydes
   (1) Lithium tri-tert-butoxyaluminum hydride, $LiAlH(O-t-Bu)_3$ is a weaker reducing agent than lithium aluminum hydride.
   (2) Reduction of an acid to an aldehyde is a two-step process: Convert the acid to the acid chloride, then reduce using lithium tri-tert-butoxyaluminum hydride.

20.14 Alkylation of Carboxylic Acids to Form Ketones 20260519

1. Carboxylic acids react with two equivalents of an organolithium reagent to give ketones.
2. Girgnard reagents cannot be used. why?
3. Mechanism

20.15 Synthesis and Use of Acid Chlorides 20260519

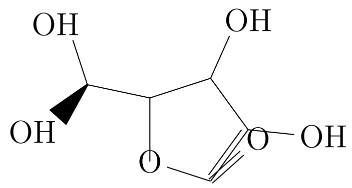
1. Halide ions are excellent leaving groups for nucleophilic acyl substitution. Acyl halides are useful intermediates for making acid derivatives.
2. Acid chlorides react with a wide range of nucleophiles.
3. The best reagents for converting carboxylic acids to acid chlorides are thionyl chloride ($SOCl_2$) and oxalyl chloride [$(COCl)_2$].
4. Mechanism
5. Acid chlorides react with alcohols to give esters through a nucleophilic acyl substitution by the addition–elimination mechanism.
6. Ammonia and amines react with acid chlorides to give amides, also through the addition–elimination mechanism of nucleophilic acyl substitution.

SUMMARY Reactions of Carboxylic Acids

Example 20.4 Arrange each group of compounds in order of increasing basicity.

Example 20.5 Arrange each group of compounds in order of increasing acidity.

Example 20.6 What do the following $\mathrm{p}K_\mathrm{a}$ values tell you about the electron-withdrawing abilities of nitro, cyano, chloro, and hydroxyl groups?

Example 20.7 Given the structure of ascorbic acid (vitamin C): 

(a) Is ascorbic acid a carboxylic acid?
(b) Compare the acid strength of ascorbic acid ($\mathrm{p}K_\mathrm{a} = 4.71$) with acetic acid.
(c) Predict which proton in ascorbic acid is the most acidic. (lose proton and compare the resonance forms)
(d) Draw the form of ascorbic acid that is present in the body (aqueous solution, $\mathrm{pH} = 7.4$).

Example 20.8 Show how you would use extractions with a separatory funnel to separate a mixture of the following compounds: benzoic acid, phenol, benzyl alcohol, aniline.

sort first.

Example 20.9 The following NMR spectra correspond to compounds of formulas (A) (B) and (C) respectively. Propose structures, and show how they are consistent with the observed absorptions.

21 Carboxylic Acid Derivatives

21.1 Introduction 20260519

1. The most important acid derivatives are esters, amides, and nitriles. Acid halides and anhydrides are also included.

21.2 Structure and Nomenclature of Acid Derivatives 20260519 20260521

21.2.1 Esters of Carboxylic Acids

1. Esters are carboxylic acid derivatives in which the hydroxyl group ($-\mathrm{OH}$) is replaced by an alkoxy group ($-\mathrm{OR}$).
2. Esters can be formed by the Fischer esterification of an acid with an alcohol.
3. The names of esters consist of two words that reflect their composite structure. The first word is derived from the *alkyl* group of the alcohol, and the second word from the *carboxylate* group of the carboxylic acid.
4. Lactones
   (1) Cyclic esters are called **lactones**.A lactone is formed from an openchain hydroxy acid in which the hydroxyl group has reacted with the acid group to form an ester.
   (2) The IUPAC names of lactones are derived by adding the term *lactone* at the end of the name of the parent acid. (open the ring and name it.)
   (3) The common names of lactones are formed by changing the *-ic* acid ending of the hydroxy acid to *-olactone*.

21.2.2 Amides

1. An amide is a composite of a carboxylic acid and ammonia or an amine.
2. Amides are only weakly basic, and we consider the amide functional group to be neutral.
3. **primary amide**, **secondary amide** or **an N-substituted amide**, **tertiary amides** or **N,N-disubstituted amides**.
4. Nomenclature: Drop the *-ic* acid or *-oic* acid suffix, and add the suffix *-amide*. For acids that are named as alkanecarboxylic acids, the amides are named by using the suffix *-carboxamide*.
5. Lactams
   (1) Cyclic amides are called **lactams**.

(2) Lactams are named like lactones, by adding the term *lactam* at the end of the IUPAC name of the parent acid.
(3) Common names of lactams are formed by changing the *-ic* acid ending of the amino acid to *-olactam*.

21.2.3 Nitriles

1. **Nitriles** contain the **cyano group**, $—C{\equiv}N$.
2. They hydrolyze to give carboxylic acids and can be synthesized by dehydration of amides.
3. Hydrolysis to an acid (H_2O), Synthesis from an acid ($POCl_3$).
4. Both the carbon atom and the nitrogen atom of the cyano group are *sp* hybridized, and the $R—C{\equiv}N$ bond angle is $180°$ (linear). The structure is similar to that of a terminal alkyne, except that the nitrogen atom has a lone pair of electrons.
5. Common names of nitriles begin with the common name of the acid, and replace the suffix *-ic acid* with the suffix *-onitrile*. The IUPAC name is constructed from the alkane name, with the suffix *-nitrile* added.

21.2.4 Acid Halides

1. The most common acyl halides are the acid chlorides (acyl chlorides).
2. An acid halide is named by replacing the *-ic acid* suffix of the acid name (either the common name or the IUPAC name) with *-yl* and the halide name. For acids that are named as alkanecarboxylic acids, the acid chlorides are named by using the suffix *-carbonyl chloride*.

21.2.5 Acid Anhydrides

1. An acid anhydride contains two molecules of an acid, with loss of a molecule of water.
2. Anhydride nomenclature: the word *acid* is changed to *anhydride* in both the common name and the IUPAC name.
3. Anhydrides composed of two different acids are called *mixed anhydrides* and are named by using the names of the individual acids.

21.2.6 Nomenclature of Multifunctional Compounds

1. acid > ester > amide > nitrile > aldehyde > ketone > alcohol > amine > alkene, alkyne
2. ethyl *o*-cyanobenzoate, 2-formylcyclohexanecarboxamide, 2-hydroxybutanenitrile
3. Summary of Functional Group Nomenclature

Example 21.1 Name the following carboxylic acid derivatives, giving both a common name and an IUPAC name where possible.

21.3 Physical Properties of Carboxylic Acid Derivatives 20260521

21.3.1 Boiling Points and Melting Points

1.

21.3.2 Solubility

21.4 Spectroscopy of Carboxylic Acid Derivatives 20260521

21.4.1 Infrared Spectroscopy

21.4.2 NMR Spectroscopy

Example 21.2 For each set of IR and NMR spectra, determine the structure of the unknown compound. Explain how your proposed structure fits the spectra.

21.5 Interconversion of Acid Derivatives by Nucleophilic Acyl Substitution 20260521 20260526

Mechanism 21.1 Addition–Elimination Mechanism of Nucleophilic Acyl Substitution

21.5.1 Reactivity of Acid Derivative

1. Acid derivatives differ greatly in their reactivity toward nucleophilic acyl substitution.
2. The order of reactivity stems partly from the basicity of the leaving groups.

21.5.2 Favorable Interconversions of Acid Derivatives

1. More reactive acid derivatives are easily converted to less reactive derivatives.

Mechanism 21.2 Conversion of an Acid Chloride to an Anhydride

This mechanism follows the standard pattern of an addition–elimination mechanism, ending with loss of a proton to give the final product.

- Step 1: Addition of the nucleophile.
- Step 2: Elimination of the leaving group.
- Step 3: Loss of a proton.

Mechanism 21.3 Conversion of an Acid Chloride to an Ester

This is another reaction that follows the standard addition–elimination mechanism, ending with loss of a proton to give the final product.

Mechanism 21.4 Conversion of an Acid Chloride to an Amide

This reaction also follows the steps of a standard addition–elimination mechanism, ending with loss of a proton to give the amide.

Mechanism 21.5 Conversion of an Acid Anhydride to an Ester

This reaction follows the standard addition–elimination mechanism, ending with loss of a proton to give the ester.

Mechanism 21.6 Conversion of an Acid Anhydride to an Amide

This reaction follows the standard addition–elimination mechanism, ending with loss of a proton to give the amide.

Mechanism 21.7 Conversion of an Ester to an Amide (Ammonolysis of an Ester)

This is yet another standard addition–elimination mechanism, ending with loss of a proton to give the amide.

21.5.3 Leaving Groups in Nucleophilic Acyl Substitutions

1. The reaction's one-step mechanism is not strongly endothermic or exothermic.
2. In the acyl substitution, the leaving group leaves in a separate second step. This second step is highly exothermic.
3. A strong base may serve as a leaving group if it leaves in a highly exothermic step, usually converting an unstable, negatively charged intermediate to a stable molecule.

Example 21.3 Which of the following proposed reactions would take place quickly under mild conditions?

21.6 Transesterification 20260526

1. Esters undergo **transesterification**, in which one alkoxy group substitutes for another, under either acidic or basic conditions.

2. An equilibrium results, and the equilibrium can be driven toward the desired ester by using a large excess of the desired alcohol or by removing the other alcohol.
3. Base-Catalyzed Transesterification
4. Acid-Catalyzed Transesterification
   (1) Consider the carbon skeletons of the reactants and products, and identify which carbon atoms in the products are likely derived from which carbon atoms in the reactants.
   (2) Consider whether any of the reactants is a strong enough electrophile to react without being activated. If not, consider how one of the reactants might be converted to a strong electrophile by protonation of a Lewis basic site.
   (3) Consider how a nucleophilic site on another reactant can attack the strong electrophile to form a bond needed in the product.
   (4) Consider how the product of nucleophilic attack might be converted to the final product or reactivated to form another bond needed in the product.
   (5) Draw out all steps of the mechanism, using curved arrows to show the movement of electrons.

Example 21.4 When ethyl 4-hydroxybutyrate is heated in the presence of a trace of a basic catalyst (sodium acetate), one of the products is a lactone. Propose a mechanism for formation of this lactone.

Mechanism 21.8 Transesterification

Following is a summary of the mechanism of transesterification under basic and acidic conditions. (Aspiring)

1. Base-catalyzed
   Base-catalyzed transesterification is a simple two-step nucleophilic acyl substitution
2. Acid-catalyzed
   Acid-catalyzed transesterification requires extra proton transfers before and after the major steps. The overall reaction takes place in two stages. The first half of the reaction involves acid-catalyzed addition of the nucleophile, and the second half involves acid-catalyzed elimination of the leaving group.

21.7 Hydrolysis of Carboxylic Acid Derivatives 20260526

21.7.1 Hydrolysis of Acid Halides and Anhydrides

1. Acid halides and anhydrides are so reactive that they hydrolyze under neutral conditions.

21.7.2 Hydrolysis of Esters

1. Acid-catalyzed hydrolysis of an ester is simply the reverse of the Fischer esterification equilibrium.
2. Basic hydrolysis of esters, called **saponification**, avoids the equilibrium of the Fischer esterification.

Mechanism 21.9 Saponification of an Ester

This is another standard addition–elimination mechanism, ending with a proton transfer to give the final products.

Example 21.5 Suppose we have some optically pure (*R*)-2-butyl acetate that has been "labeled" with the heavy ^{18}O isotope at one oxygen atom as shown.

21.7.3 Hydrolysis of Amides

1.

Mechanism 21.10 Basic Hydrolysis of an Amide

Mechanism 21.11 Acidic Hydrolysis of an Amide

This mechanism takes place in two stages.

21.7.4 Hydrolysis of Nitriles

1. Nitriles are hydrolyzed to amides, and further to carboxylic acids, by heating with aqueous acid or base.
2. Mild conditions can hydrolyze a nitrile only as far as the amide. Stronger conditions can hydrolyze it all the way to the carboxylic acid.

Mechanism 21.12 Base-Catalyzed Hydrolysis of a Nitrile

Example 21.6 The mechanism for acidic hydrolysis of a nitrile resembles the basic hydrolysis, except that the nitrile is first protonated, activating it toward attack by a weak nucleophile (water). Under acidic conditions, the proton transfer (tautomerism) involves protonation on nitrogen followed by deprotonation on oxygen. Propose a mechanism for the acid-catalyzed hydrolysis of benzonitrile to benzamide.

21.8 Reduction of Acid Derivatives 20260526

21.8.1 Reduction to Alcohols

1. Lithium aluminum hydride reduces acids, acid chlorides, anhydrides, and esters to primary alcohols.

Mechanism 21.13 Hydride Reduction of an Ester

21.8.2 Reduction to Aldehydes

21.8.3 Reduction to Amines

1.

21.9 Reactions of Acid Derivatives with Organometallic Reagents 20260526

1. Esters and Acid Chlorides
   (1) Grignard and organolithium reagents add twice to acid chlorides and esters to give alkoxides (Section 10-9D). Protonation of the alkoxides gives alcohols.
2. Nitriles
   (1) A Grignard or organolithium reagent attacks the electrophilic cyano group to form the salt of an imine.
   (2) Acidic hydrolysis of the salt (in a separate step) gives the imine, which is further hydrolyzed to a ketone (Section 18-9).

Example 21.7 Acidic hydrolysis of the salt (in a separate step) gives the imine, which is further hydrolyzed to a ketone (Section 18-9).

21.10 Summary of the Chemistry of Acid Chlorides 20260526

1. Synthesis of Acid Chlorides: Acid chlorides (acyl chlorides) are synthesized from the corresponding carboxylic acids, $SOCl_2$ and $COCl_2$ most convenient.
2. Reactions of Acid Chlorides
3. Friedel–Crafts Acylation of Aromatic Rings

21.11 Summary of the Chemistry of Anhydrides 20260526

1. 1

2. Acetic Anhydride: The most common industrial synthesis begins by dehydrating acetic acid to give ketene.
3. General Anhydride Synthesis: The most general method for making anhydrides is the reaction of an acid chloride with a carboxylic acid or a carboxylate salt.
4. Reactions of Anhydrides: Like acid chlorides, anhydrides participate in the Friedel–Crafts acylation. Cyclic anhydrides can provide additional functionality on the side chain of the aromatic product.
5. Three specific instances when anhydrides are preferred.
   (1)

Example 21.8 (a) Give the products expected when acetic formic anhydride reacts with (i) aniline and (ii) benzyl alcohol.

(b) Propose mechanisms for these reactions.

21.12 Summary of the Chemistry of Esters 20260526

1. Synthesis of Esters
2. Reactions of Esters
3. Formation of Lactones

21.13 Summary of the Chemistry of Amides 20260526

1. Synthesis of Amides
2. Reactions of Amides
3. Dehydration of Amides to Nitriles: Strong dehydrating agents can remove the elements of water from a primary amide to give a nitrile.
4. Formation of Lactams
5. Biological Reactivity of β-Lactams: penicillin
6. Polyamides: Nylon

21.14 Summary of the Chemistry of Nitriles 20260528

1. Synthesis
2. Reactions of Nitriles

21.15 Thioesters 20260528

1. A **thioester** is formed from a carboxylic acid and a thiol.
2. Thioesters are more reactive toward nucleophilic acyl substitution than normal esters, but less reactive than acid chlorides and anhydrides.
3. The overlap is weak and relatively ineffective.

21.16 Esters and Amides of Carbonic Acid 20260528

1. **Carbonic acid** is formed reversibly whenever carbon dioxide dissolves in water.
2. **Carbonate esters** are diesters of carbonic acid, with two alkoxy groups replacing the hydroxyl groups of carbonic acid.
3. **Ureas** are diamides of carbonic acid, with two nitrogen atoms bonded to the carbonyl group.
4. **Carbamate esters** (**urethanes**) are the stable esters of the unstable **carbamic acid**, the monoamide of carbonic acid.
5. Many of these derivatives can be synthesized by nucleophilic acyl substitution from phosgene, the acid chloride of carbonic acid. Another way of making urethanes is to treat an alcohol or a phenol with an **isocyanate**, which is an anhydride of a carbamic acid.
6. Polycarbonates and Polyurethanes
   (1) two large classes of polymers are bonded by linkages containing these functional groups: the **polycarbonates** and the **polyurethanes**.
   (2) *bisphenol A*

Example 21.9 An unknown compound gives a mass spectrum with a weak molecular ion at *m/z* 113 and a prominent ion at *m/z* 68. Its NMR and IR spectra are shown here. Determine the structure, and show how it is consistent with the observed absorptions. Propose a favorable fragmentation to explain the prominent MS peak at *m/z* 68.

SOLUTION

MS 113-odd-N 113 - 68 = 45

IR 3000 medium $-NH_2$?, 2200 triple bond, 1750 carbonyl, 1200 $C-O$

C NMR 5 C, 5 * 12 = 60

till now, C5 N O2 sum = 106, so 7H

H NMR one single $-CH_2$, one $-CH_2$ and $-CH_3$ together (split).

$N\equiv C-CH_2-\overset{\overset{\displaystyle O}{\|}}{C}-O-CH_2-CH_3$ $-OEt = 45$

Example 21.10 An unknown compound gives the NMR, IR, and mass spectra shown next. Propose a structure, and show how it is consistent with the observed absorptions. Show fragmentations that account for the prominent ion at **m/z** 69 and the smaller peak at **m/z** 99.

SOLUTION

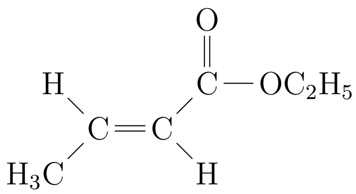
MS 114 - 69 = 45($-OEt$), 114 - 99 = 13($-CH_3$), no benzyl(91)

IR 3000 no $-OH$ $-NH_2$ 1700-1600 $C=C-\overset{\overset{\displaystyle O}{\|}}{C}$ classic

C NMR 5 C? 6?(because of 5 kind of H)

H NMR 6-7 H of $C=C$ beside it is $-CH_3$, the rest is $-OEt$.

notice: E not Z



Example 21.11 The IR spectrum, ^{13}C NMR spectrum, and ^{1}H NMR spectrum of an unknown compound ($C_6H_8O_3$) appear next. Determine the structure, and show how it is consistent with the spectra.

SOLUTION

Ω = 3, 2 carbonyl and 1 ring? no anhydride(1800)

C NMR one over 200

H NMR 3:1:1:1:2 the highest one not splitted, $-CH_3$ to $C=O$.

Example 21.12 An unknown compound of molecular formula C_5H_9NO gives the IR and NMR spectra shown here. The broad NMR peak at $\delta 7.55$ disappears when the sample is shaken with D_2O. Propose a structure, and show how it is consistent with the absorptions in the spectra.

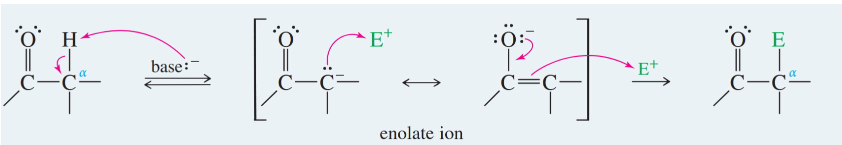
22 Condensations and Alpha Substitutions of Carbonyl Compounds

22.1 Introduction 20260528

1. **Alpha** (α) **substitutions** replace a hydrogen atom at the α **carbon atom** by some other group.
2. Alpha substitutions generally take place when the carbonyl compound is converted to its enolate ion or its enol tautomer.

Mechanism 22.1 Alpha Substitution

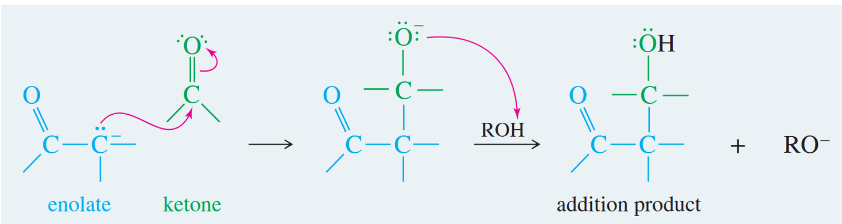
- ***Step 1:*** Deprotonation of a carbon to form an enolate.
- ***Step 2:*** Nucleophilic attack on an electrophile.



3. Carbonyl **condensations**: alpha substitutions where the electrophile is another carbonyl compound.
4. If the electrophile is a ketone or an aldehyde: nucleophilic addition.

Mechanism 22.2 Addition of an Enolate to Ketones and Aldehydes (a Condensation)

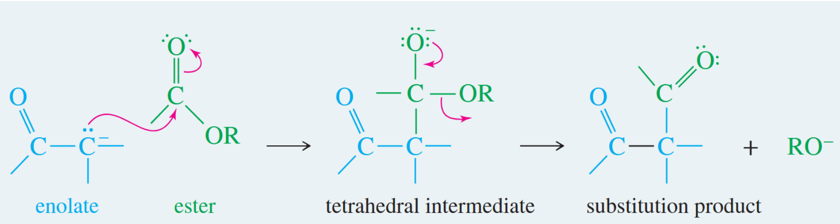
- ***Step 1:*** The enolate adds to the carbonyl group.
- ***Step 2:*** Protonation of the alkoxide.



5. If the electrophile is an ester: nucleophilic acyl substitution.

Mechanism 22.3 Substitution of an Enolate on an Ester (a Condensation)

- ***Step 1:*** Addition of the enolate.
- ***Step 2:*** Elimination of the alkoxide.



6. Alpha substitutions and condensations of carbonyl compounds are some of the most common methods for forming carbon-carbon bonds.
7. Considering the structure and formation of enols and enolate ions.